

Power Centers and Hull-Power Centers

Attractivity, Exact Certificates, and Dimension Dependence

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Revised collection of results through 6 October 2026

Abstract

We collect the results obtained for discrete power centers and their uniform-solid simplex counterparts, called hull-power centers. For discrete power centers, the universal vertex-attractivity interval is exactly $[4 - 2\sqrt{2}, 4 + 2\sqrt{2}]$; every exponent outside this interval has a proper triangle counterexample. Universal atomic attractivity transfers to hull-power centers by empirical approximation and successive parallel material motions. Uniform barycentric mass has additional structure: an analytic argument extends the common positive interval, in every intrinsic simplex dimension, to $4 + 2\sqrt{2} + 1/2,000,000$. For triangles, complete exact integer interval certificates prove every real exponent in $[4 - 2\sqrt{2}, 16]$ and the isolated exponents 20 and 21, with explicit finite-motion constants. The intervening interval is not yet certified. At 65/64 we have local hull certificates and an atomic counterexample, but no all-shape hull theorem. Exact finite failures occur at 21.64, 22, and 30 for triangles, at 20 for tetrahedra, at 18 in dimension 12, and at 16 in dimension 232, improving the earlier bound 302. A numerical family of failure thresholds decreases toward approximately 15.8468 as dimension grows; global minimality is unproved. New structural results give an exact disk-containment test, equal-edge-moment equations, and unique centered projective families. Every nonregular such family eventually fails, whereas one fixed triangle and one fixed finite motion lose and regain attractivity. Thus fixed-shape monotonicity is false, while all-shape downward closure remains open. A new ETC fingerprint comparison at 33 selected powers finds 82 distinct neighbors: 33 have exact finite failures, three have positive proofs, and 46 remain unclassified. Attractive targets can have failing neighbors, and failing targets can have attractive neighbors. We separate analytic proofs, completed computer-assisted certificates, numerical evidence, and unresolved claims.

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1 Definitions and the current results

Let $\mathbf{P} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a finite labelled list in $\mathbb{R}^D$. The integer D denotes the **ambient** dimension. A proper simplex with n vertices has **intrinsic** dimension $d = n - 1$, possibly with $D > d$.

Definition 1.1 (The two power-center families). For $p > 1$, the discrete or atomic power center is

$$\mathbf{M}_p(\mathbf{P}) = \arg \min_{x \in \mathbb{R}^D} \frac{1}{p} \sum_{i=1}^n \|\mathbf{x} - \mathbf{v}_i\|^p. \quad (1)$$

Let $\lambda \in \text{Dirichlet}(1, \dots, 1)$ on the fixed reference simplex $\Lambda_n = \{\lambda_i \geq 0 : \sum_i \lambda_i = 1\}$. The hull-power center is

$$\mathbf{U}_p(\mathbf{P}) = \arg \min_{x \in \mathbb{R}^D} \frac{1}{p} \mathbb{E} \|\mathbf{x} - \mathbf{Y}_P\|^p, \quad \mathbf{Y}_P = \sum_i \lambda_i \mathbf{v}_i. \quad (2)$$

For a proper simplex $\mathbf{S} = \text{conv } \mathbf{P}$, this is equivalently the minimizer of

$$\frac{1}{p \text{Vol}_d(\mathbf{S})} \int_{\mathbf{S}} \|\mathbf{x} - \mathbf{y}\|^p d\mathbf{y}. \quad (3)$$

Thus the hull model distributes uniform mass throughout the solid simplex.

The probability normalization and the factor $1/p$ do not change the minimizer at a fixed proper simplex. They make the measure interpretation and its limiting convention precise. The word “hull” here always refers to (2); no uniform-solid prescription for general nonsimplicial polytopes is asserted.

Write $\mathbf{P}_{i,h}$ for the list obtained by replacing $\mathbf{v}_i$ with $\mathbf{v}_i + \mathbf{h}$.

Definition 1.2 (Finite vertex attractivity). A rule $\mathbf{X}$ is vertex-attractive if

$$\mathbf{h} \cdot \left(\mathbf{X}(\mathbf{P}_{i,h}) - \mathbf{X}(\mathbf{P}) \right) \geq 0 \quad (4)$$

for every configuration, moved vertex, and displacement allowed in the stated domain. At a differentiable configuration, its local condition is

$$\text{Sym } \mathbf{D}_{v_i} \mathbf{X} \succeq 0, \quad \text{Sym } \mathbf{A} = (\mathbf{A} + \mathbf{A}^T)/2. \quad (5)$$

This property concerns the response to moving input data. Convexity of the optimization problem alone does not establish it. Integrating $\mathbf{h}^T \mathbf{D}_{v_i} \mathbf{X}(\mathbf{P}_{i,th}) \mathbf{h} \geq 0$ gives the finite law on regular paths; continuity joins the degenerate configurations treated below.

Put

$$p_- = 4 - 2\sqrt{2}, \quad p_+ = 4 + 2\sqrt{2}, \quad \delta = \frac{1}{2,000,000}.$$

The following table states the present positive conclusions. “Every shape” includes the fixed barycentric extension at degeneracies.

Strict proper finite failures for $\mathbf{U}_p$ are recorded in Table 3. In particular, positivity at 16 for every triangle and failure at 16 in intrinsic dimension 232 already prove different behavior among intrinsic dimensions $d \geq 2$.

2 Measures, continuity, and elementary exact cases

2.1 Existence, invariance, and the degeneracy convention

For a compactly supported probability measure μ , write

$$\mathbf{M}_p(\mu) = \arg \min_x \frac{1}{p} \int \|\mathbf{x} - \mathbf{y}\|^p d\mu(\mathbf{y}).$$

Family and scope	Proved exponents	Nature of result
M_p , all finite lists	$[p_-, p_+]$, exactly	Analytic classification
U_p , every simplex dimension	$[p_-, p_+ + \delta]$	Analytic sufficient interval
U_p , every triangle	$[p_-, 16]$	Analytic and exact interval proof
U_p , every triangle	$\{20, 21\}$	Separate exact endpoint proofs
M_p, U_p , two vertices	Every $p > 1$	Midpoint identity

Table 1: Universal positive results as of 5 October 2026. All entries allow arbitrary finite ambient Euclidean embedding dimension. The continuous interval ends at 16; the separate endpoints 20 and 21 do not fill the intervening gap. No sharp hull endpoint is claimed.

The cost is coercive and strictly convex for $p > 1$, so the minimizer exists and is unique. Projection onto the closed convex hull of the support decreases every distance from an exterior point, proving internality. Both families are equivariant under translations, orthogonal transformations, and positive scalings. Equal vertex weights or uniform barycentric weights also give permutation invariance. Arbitrary affine equivariance is not claimed.

If $\mathbf{v}$ is uniform probability on Λ_n and $T_P(\lambda) = \sum_i \lambda_i \mathbf{v}_i$, then $U_p(\mathbf{P}) = M_p((T_P)_\# \mathbf{v})$. We retain this same reference law when vertices become dependent or coincide. It provides a continuous extension; resetting the measure to intrinsic uniform volume on a collapsed hull gives a different model. For example, a triangle flattened onto coordinates $0, 1, 3$ has quadratic barycentric center $4/3$, whereas uniform length on $[0, 3]$ has quadratic center $3/2$. An unnormalized two-dimensional integral on that collapsed hull is zero and cannot define a unique center.

Proposition 2.1 (Compact empirical stability). If $\mu_j \sqsubseteq \mu$ and all measures have support in one compact set, then $M_p(\mu_j) \rightarrow M_p(\mu)$. Convergence is uniform in p on every compact interval $I \sqsubseteq (1, \infty)$. Consequently

$$d_H(\text{conv}\{M_p(\mu_j) : p \sqsubseteq I\}, \text{conv}\{M_p(\mu) : p \sqsubseteq I\}) \rightarrow 0.$$

Proof. All minimizers lie in one compact convex hull. Joint continuity of $\|\mathbf{x} - \mathbf{y}\|^p$, including $\mathbf{x} = \mathbf{y}$, and a finite net in $(p, \mathbf{x})$ give uniform convergence of the costs there. Any limit of minimizers minimizes the limit cost, and uniqueness identifies it. A hypothetical failure of uniformity admits a convergent exponent subsequence and contradicts this argument. Pairing identical exponents in convex combinations bounds the Hausdorff distance by the supremum of the center errors. $\square$

This is convergence of the **power-generated** regions, rather than a theorem about convergence of the full class of attractive center rules.

2.2 Centroids, segments, symmetry, and quartic centers

At $p = 2$ both families equal $\mathbf{G} = n^{-1} \sum_i \mathbf{v}_i$, and

$$h \cdot \sum U_2(\mathbf{P}_{i,h}) - U_2(\mathbf{P}) = \sum h^2/n. \quad (6)$$

For two equal-weight vertices, M_p is their midpoint for all $p > 1$. The uniform segment center is also the midpoint; the latter remains the unique minimizer for every $p > 0$. Indeed, in intrinsic coordinates $\mathbf{a} \leq \mathbf{x} \leq \mathbf{b}$, its unnormalized integral equals $[(\mathbf{x} - \mathbf{a})^{p+1} + (\mathbf{b} - \mathbf{x})^{p+1}]/(p+1)$ and is minimized at $(\mathbf{a} + \mathbf{b})/2$. Central symmetry forces the center to be its symmetry point. In particular, on a regular simplex $M_p = U_p = \mathbf{G}$ for every $p > 1$. Positivity at a symmetric shape or along a restricted symmetry family does not classify unrestricted vertex motions on all shapes.

The quartic case admits an efficient exact moment reduction. Set $\mathbf{z}_i = \mathbf{v}_i - \mathbf{G}$, and for uniform simplex mass put

$$\Sigma = \frac{1}{n(n+1)} \sum_i \mathbf{z}_i \mathbf{z}_i^T, \quad \mathbf{m}_2 = \text{tr } \Sigma, \quad \mathbf{m}_3 = \frac{2}{n(n+1)(n+2)} \sum_i \|\mathbf{z}_i\|^2 \mathbf{z}_i. \quad (7)$$

Writing $\mathbf{U}_4 = \mathbf{G} + \mathbf{c}$, expansion of its force equation gives

$$[(\|\mathbf{c}\|^2 + \mathbf{m}_2)\mathbf{I} + 2\Sigma]\mathbf{c} = \mathbf{m}_3. \quad (8)$$

If $\mathbf{m}_3 = 0$, then $\mathbf{c} = 0$. Otherwise the scalar equation

$$t = \left[(t + \mathbf{m}_2)\mathbf{I} + 2\Sigma \right]^{-1} \mathbf{m}_3 \|\mathbf{c}\|^2, \quad t = \|\mathbf{c}\|^2 \geq 0,$$

has a unique root, since its right side decreases while its left side increases. It is bracketed by 0 and $\left[\mathbf{m}_2\mathbf{I} + 2\Sigma \right]^{-1} \mathbf{m}_3 \|\mathbf{c}\|^2$. For the atomic center, the same general moment equation holds with $\Sigma = n^{-1} \sum_i \mathbf{z}_i \mathbf{z}_i^T$ and $\mathbf{m}_3 = n^{-1} \sum_i \|\mathbf{z}_i\|^2 \mathbf{z}_i$. These different moments explain why the two centers generally disagree away from quadratic or symmetry-forced cases.

Prior work also proves that the entire atomic family $\mathbf{M}_p$, $1 < p < \infty$, is attractive on every regular-base axial simplex when deformations are restricted to its one-parameter symmetry family [1]. This restricted-motion result does not classify arbitrary vertex motions.

3 The exact atomic classification

3.1 Stiffness and the operator-angle mechanism

At a configuration whose atomic center $\mathbf{x}$ is distinct from the anchors, let $\mathbf{r}_i = \|\mathbf{x} - \mathbf{v}_i\|$, $\mathbf{e}_i = (\mathbf{x} - \mathbf{v}_i)/r_i$, and $k = p - 1$. Then

$$\mathbf{K}_i = r_i^{p-2} [\mathbf{I} + (p-2)\mathbf{e}_i \mathbf{e}_i^T], \quad \mathbf{K} = \sum_i \mathbf{K}_i, \quad \mathbf{D}_{v_i} \mathbf{M}_p = \mathbf{K}^{-1} \mathbf{K}_i. \quad (9)$$

Positive atom masses simply multiply the corresponding $\mathbf{K}_i$. Each $\mathbf{K}_i$ is positive definite, with radial-to-tangential eigenvalue ratio k . The product $\mathbf{K}^{-1} \mathbf{K}_i$ has positive response in the Hessian metric, because $\mathbf{K}(\mathbf{K}^{-1} \mathbf{K}_i) = \mathbf{K}_i$. Its Euclidean symmetric part can nevertheless be indefinite.

If an operator $\mathbf{A} \neq 0$ has eigenvalues in $[m, M]$, then

$$\frac{\mathbf{h}^T \mathbf{A} \mathbf{h}}{\|\mathbf{h}\|^2} \geq \frac{2\sqrt{mM}}{m+M}. \quad (10)$$

For a unit vector, regard its squared eigenvector coordinates as a probability law on eigenvalues $\mathbf{s} \in [m, M]$. Writing $\mathbf{a} = \mathbb{E}\mathbf{s}$, the inequality $(\mathbf{s} - m)(\mathbf{s} - M) \leq 0$ gives $\mathbb{E}\mathbf{s}^2 \leq (m+M)\mathbf{a} - mM$; completing a square yields (10). Thus a condition number at most q permits a turning angle at most $\varphi(q) = \arccos(2\sqrt{q}/(1+q))$.

All $\mathbf{K}_i$ and their positive sum have condition number at most $q = \max(k, 1/k)$. The same holds for $\mathbf{K}^{-1}$. Hence

$$\mathbf{h}^T \mathbf{K}^{-1} \mathbf{K}_i \mathbf{h} = (\mathbf{K}^{-1} \mathbf{h}) \cdot (\mathbf{K}_i \mathbf{h}) \geq 0 \quad \text{if} \quad 2\varphi(q) \leq \pi/2,$$

which is exactly $q \leq 3 + 2\sqrt{2}$. More generally, the configuration-dependent condition

$$\varphi(\kappa(\mathbf{K})) + \varphi(\kappa(\mathbf{K}_i)) \leq \pi/2$$

certifies an atomic local response. For the hull response one replaces $\mathbf{K}_i$ by $\mathbf{L}_i$. Such a test can certify particular shapes outside a universal band without proving an all-shape theorem.

Theorem 3.1 (Exact atomic power interval; [1, 2]). The following are equivalent for $p > 1$: M_p satisfies the finite law (4) for every finite Euclidean configuration; it satisfies that law for every triangle; and

$$p_- \leq p \leq p_+. \quad (11)$$

Outside this band, noncollinear three-point counterexamples suffice.

For collisions and finite motions, replace r^p/p by $(r^2 + \varepsilon^2)^{p/2}/p$. Its stiffness ratio is $1 + (p - 2)r^2/(r^2 + \varepsilon^2)$, between 1 and $p - 1$. The same angle bound applies everywhere. Integrating the regularized response and passing to the limit by compact uniform convergence proves sufficiency, including collisions. In the open atomic band, every regular finite configuration has strictly positive symmetric responses. Likewise, strict pointwise elasticity bounds give strictness at each fixed regular radial configuration, since only finitely many radii occur; a uniform gap is needed for a configuration-uniform quantitative bound.

3.2 Why three vertices suffice outside the band

Prescribe nonzero force vectors $\mathbf{f}_i$ with $\sum_i \mathbf{f}_i = 0$, and realize them by anchors $\mathbf{v}_i = -\|\mathbf{f}_i\|^{1/k-1} \mathbf{f}_i$. Their center is exactly the origin. Put $\mathbf{d} = (1, 0)^T$, $\mathbf{e} = (1, 1)^T / \sqrt{2}$, and

$$\mathbf{D}_k = \begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{B}_k = \frac{1}{2} \begin{pmatrix} k+1 & k-1 \\ k-1 & k+1 \end{pmatrix}, \quad \beta = (k-1)/k.$$

For $k > 3 + 2\sqrt{2}$, use $(\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3) = \sqrt{\varepsilon}(\mathbf{d} - \varepsilon\mathbf{e}/2, -\mathbf{d} - \varepsilon\mathbf{e}/2, \varepsilon\mathbf{e})$. As $\varepsilon \downarrow 0$, $\varepsilon^{-\beta} \text{Sym } \mathbf{D}_{v_3} M_p \rightarrow \text{Sym}(\mathbf{D}_k^{-1} \mathbf{B}_k)/2$. For $0 < k < 3 - 2\sqrt{2}$, use $(\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3) = (\mathbf{e}, -\mathbf{e} - \varepsilon\mathbf{d}, \varepsilon\mathbf{d})$; then $\varepsilon^\beta \text{Sym } \mathbf{D}_{v_1} M_p$ has the same limit. The limiting matrix $\mathbf{H} = \text{Sym}(\mathbf{D}_k^{-1} \mathbf{B}_k)$ satisfies

$$\det \mathbf{H} = -\frac{(k+1)^2(k^2 - 6k + 1)}{16k^2} < 0 \quad (12)$$

outside the band. For sufficiently small positive ε the constructed triangles are proper and the specified response is indefinite. Differentiability turns a negative local direction into a finite failure. The supplied rational interval certificates give concrete instances at $p = 7$, $\varepsilon = 1/10000$, moving vertex 3, and $p = 23/20$, $\varepsilon = 1/5$, moving vertex 1.

3.3 The broader radial theory

For completeness, the original classification extends to $X_\psi(\mathbf{P}) = \arg \min_x \sum_i \psi(\|\mathbf{x} - \mathbf{v}_i\|)$. Assume ψ is increasing, strictly convex, coercive, and $\mathbf{C}^2$ away from zero, with $\mathbf{F} = \psi' > 0$ and $\mathbf{F}' = \psi'' > 0$ there. Its elasticity is $\eta(\mathbf{r}) = \mathbf{r} \mathbf{F}'(\mathbf{r}) / \mathbf{F}(\mathbf{r})$ and its individual stiffness is

$$\mathbf{K}_i = \frac{\mathbf{F}(\mathbf{r}_i)}{\mathbf{r}_i} [\mathbf{I} + (\eta(\mathbf{r}_i) - 1) \mathbf{e}_i \mathbf{e}_i^T].$$

Theorem 3.2 (Radial classification and scale rigidity; [1]). The universal finite atomic attractiveness law holds exactly when

$$3 - 2\sqrt{2} \leq \eta(\mathbf{r}) \leq 3 + 2\sqrt{2} \quad (\mathbf{r} > 0). \quad (13)$$

Under the same hypotheses, similarity equivariance for every finite list holds exactly for $\mathbf{F}(\mathbf{r}) = C r^k$, $C > 0$, $k > 0$; these are the pure power laws.

The angle proof gives sufficiency. Necessity at one radius $\mathbf{r}_0$ uses m antipodal pairs on the $\mathbf{d}$ -axis and one pair on the $\mathbf{e}$ -axis, all at radius $\mathbf{r}_0$. Force cancellation fixes the origin, and the

moved $\mathbf{e}$ -anchor has response $(\mathbf{mD}_k + \mathbf{B}_k)^{-1}\mathbf{B}_k/2$, with $\mathbf{k} = \eta(\mathbf{r}_0)$. Its positively rescaled limit is $\mathbf{H}/2$ from (12). An explicit sufficient number of pairs is

$$m > \frac{4k((k+1) + \sqrt{2|k-1|})}{(k+1)(k^2 - 6k + 1)} \quad \text{when } k^2 - 6k + 1 > 0.$$

Nearby distinct antipodal pairs preserve a strict negative sign. A log-force regularization, quadratic near zero and preserving (13), supplies the collision case. Scale rigidity follows from two-cluster equilibria, which force $\mathbf{F}(\mathbf{s}\mathbf{r})/\mathbf{F}(\mathbf{r})$ to depend only on $\mathbf{s}$; continuity and multiplicativity make it $\mathbf{s}^k$.

The classification has three further consequences already proved in the original theory. First, it is equivalent to the growth inequalities

$$(\mathbf{s}/\mathbf{r})^{3-2\sqrt{2}} \leq \mathbf{F}(\mathbf{s})/\mathbf{F}(\mathbf{r}) \leq (\mathbf{s}/\mathbf{r})^{3+2\sqrt{2}} \quad (0 < \mathbf{r} < \mathbf{s}).$$

In particular, a universally attractive force cannot saturate at infinity. Second, a finite positive mixture $\Psi(\mathbf{r}) = \sum_j \mathbf{c}_j \mathbf{r}^{\mathbf{p}_j} / \mathbf{p}_j$, $\mathbf{c}_j > 0$, is universally attractive exactly when every $\mathbf{p}_j \in [\mathbf{p}_-, \mathbf{p}_+]$: its elasticity is the weighted average of $\mathbf{p}_j - 1$, and its smallest and largest exponents dominate at zero and infinity. Unless all exponents coincide, it is not similarity equivariant. Positive integral mixtures have the sufficient implication whenever differentiation is legitimate. Third, distinct labelled radial interactions whose elasticities all lie in the band have attractive local responses and the finite law on collision-free regular paths. This last statement concerns a labelled system and does not assert permutation invariance of an unlabelled center.

4 From atomic centers to hull-power centers

4.1 Material response and the transfer theorem

Under a smooth-kernel or appropriate dominated-differentiation hypothesis, the push-forward motion $\mathbf{y} \mapsto \mathbf{y} + \mathbf{t}\mathbf{v}(\mathbf{y})$ gives

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{K}^{-1} \int \mathbf{H}_p(\mathbf{x} - \mathbf{y}) \mathbf{v}(\mathbf{y}) \, d\mu(\mathbf{y}), \\ \mathbf{H}_p(\mathbf{w}) &= \mathbf{w} \mathbf{w}^{p-2} \mathbf{I} + (\mathbf{p} - 2) \mathbf{w} \mathbf{w}^{p-4} \mathbf{w} \mathbf{w}^T, \quad \mathbf{K} = \int \mathbf{H}_p(\mathbf{x} - \mathbf{y}) \, d\mu(\mathbf{y}). \end{aligned} \quad (14)$$

For $\mathbf{p} \geq 2$ the kernel has its continuous extension at zero. A separated-support hypothesis also suffices, but does not cover an interior singularity in a uniformly filled solid when $1 < \mathbf{p} < 2$. The following finite argument avoids this restriction.

Theorem 4.1 (Universal atomic-to-hull transfer). *If a fixed exponent has the universal finite atomic attractivity law, then its hull-power center is vertex-attractive for every simplex. The conclusion also holds for any fixed probability law on Λ_n , independent of $\mathbf{P}$. Consequently all hull-power centers are attractive for $\mathbf{p} \in [\mathbf{p}_-, \mathbf{p}_+]$.*

Proof. Approximate the reference probability by equally weighted empirical atoms λ^j . A vertex motion $\mathbf{h}$ moves the physical atom $\sum_k \lambda_k^j \mathbf{v}_k$ by $\lambda_i^j \mathbf{h}$. Move these atoms one at a time. If $\lambda_i^j > 0$, the atomic inequality divided by λ_i^j says $\mathbf{h} \cdot (\mathbf{x}_{j+1} - \mathbf{x}_j) \geq 0$; if it is zero, the atom does not move. Summation telescopes. Both endpoint empirical measures converge weakly, with common compact support. Proposition 2.1 passes the finite inequality to the limiting centers. The same proof uses arbitrary positive-weight approximations because the atomic sufficiency proof permits fixed positive weights. $\square$

This proves the precise implication asked about the two models. It needs the atomic law for the approximating finite clouds, not merely positivity of $\mathbf{M}_p$ on the vertices of a single simplex.

Moving a vertex of a solid moves a positive-mass collection of material points; moving one individual point of a nonatomic measure changes no mass. More generally, bounded nonnegative multiples $c(y)h$ admit the same approximation argument.

Corollary 4.2 (Failure of the converse). **Hull-power attractivity does not imply universal atomic power attractivity.**

Proof. Theorem 6.1 below proves U_{16} attractive on every triangle; Theorem 3.1 gives proper atomic triangle failures at 16. Even an implication from hull attractivity in **every** simplex dimension to universal atomic attractivity is false: Theorem 5.1 proves the former at $p_+ + \delta$, strictly beyond the exact atomic band. $\square$

4.2 Response, internality, and analytic regularity of the solid

For a proper simplex, its hull-power center is strictly interior. Indeed, stationarity expresses it as a barycenter of the solid with positive density proportional to $|x - y|^{p-2}$; this scalar weight is integrable since $d + p - 2 > 0$. Every supporting facet has a positive inward distance average. Differentiating the fixed reference integral yields

$$D_{v_i} U_p = K^{-1} L_i, \quad K = \mathbb{E} H_p(x - Y_P), \quad L_i = \mathbb{E} [\lambda_i H_p(x - Y_P)]. \quad (15)$$

Both matrices are positive definite on the intrinsic span. Their sum identity $\sum_i L_i = K$ gives $\sum_i D_{v_i} U_p = I$, as required by translations. Replacing L_i by the atomic tensor at v_i would change the problem. Near an interior point the kernel has size r^{p-2} ; its integral over a ball of radius ε is $O(\varepsilon^{d+p-2})$. Removing this ball and then taking its limit justifies differentiation for $p > 1$ on proper simplices, including the singular range $1 < p < 2$.

Proposition 4.3 (Analyticity on proper simplices; [2]). **For each fixed $p > 1$, both M_p and U_p are real-analytic vertex maps on the proper-simplex domain.**

Proof. An atomic minimizer cannot be a vertex of a proper simplex: the other independent edge vectors cannot have a vanishing positive combination. The ordinary analytic implicit function theorem applies away from anchors. For the solid center, let F run over the facets of S , with outward normal n_F and positive distance $h_F(x)$ from an interior x . Divergence gives

$$\int_S |y - x|^{p-2} dy = \frac{1}{d+p} \sum_F h_F(x) \int_F |y - x|^{p-2} dy. \quad (16)$$

The flux of a removed small ball tends to zero. Every facet remains separated from x on a small proper-parameter neighborhood. A fixed-facet parametrization makes its area factor and integrand real analytic there. Thus the cost is analytic near its interior minimizer, with positive definite Hessian, and the analytic implicit function theorem applies. Local analytic coordinates on the intrinsic span give the same conclusion in a larger ambient space. $\square$

No blanket analyticity is asserted for arbitrary dependent atomic lists: their noninteger-power centers may coincide with atoms.

For triangles there is also a useful boundary-only representation. Put $w = y - x$, $F_p(w) = |w|^{p-2}w$, and use unnormalized area integrals $m = \int_S F_p(y - x) dy$, $K = \int_S D F_p(y - x) dy$, and $L_i = \int_S \lambda_i(y) D F_p(y - x) dy$. For each edge E let n_E be its outward normal and $h_E = (y - x) \cdot n_E$. Then

$$\begin{aligned} m &= \frac{1}{p+1} \sum_E h_E \int_E F_p(y - x) ds, \\ K &= \sum_E \int_E F_p(y - x) n_E^T ds, \\ L_i &= \sum_E \int_E \lambda_i(y) F_p(y - x) n_E^T ds - m \lambda_i^T. \end{aligned} \quad (17)$$

These follow by divergence and integration by parts. The last correction is required off stationarity and vanishes at the center. Since the center is interior, edge integrands are separated from its singular point. These formulas support the low-power numerical evaluator and the rational noninteger failure certificate.

5 A dimension-independent improvement beyond the atomic band

The atomic operator bound permits independently oriented stiffness tensors. Uniform barycentric mass imposes correlations between the total directional stiffness and each vertex-weighted stiffness. Those correlations produce the first strict extension beyond p_+ .

Theorem 5.1 (Common hull-power interval; [4]). **For every simplex dimension and every finite ambient embedding dimension,**

$$\boxed{p_- \leq p \leq p_+ + \frac{1}{2,000,000}} \quad (18)$$

implies finite vertex attractivity of U_p , including the fixed barycentric extension. The added interval above p_+ has strictly positive local response at every nonconstant configuration.

5.1 An exponential moment estimate

Put $t = p - 2$, $k = 1 + t$, $w = Y_P - x$, and define

$$T = \mathbb{E}[w^t], \quad a_i = \mathbb{E}[\lambda_i w^t] / T, \\ Q = \frac{\mathbb{E}[w^t w w^T]}{T}, \quad Q_i = \frac{\mathbb{E}[\lambda_i w^t w w^T]}{a_i T}.$$

For a nonconstant list in the powers considered here, $T, a_i > 0$ and $Q, Q_i \geq 0$ have trace one. The response is

$$J_i = a_i A^{-1} B_i, \quad A = I + tQ, \quad B_i = I + tQ_i. \quad (19)$$

If X_j are independent mean-one exponential random variables and $S = \sum_j X_j$, then $\lambda_j = X_j/S$ is uniform barycentric and independent of S . For $Z = \sum_j X_j(v_j - x) = Sw$, the direction matrices become

$$Q = \frac{\mathbb{E}[Z^t Z Z^T]}{\mathbb{E}[Z^t]}, \quad Q_i = \frac{\mathbb{E}[X_i Z^t Z Z^T]}{\mathbb{E}[X_i Z^t]}.$$

Lemma 5.2. For $2 \leq t \leq 5$, a mean-one exponential variable X , fixed vectors a, b , and $R = \mathbb{E}[b + X a^t]$,

$$R/32 \leq \mathbb{E}[X b + X a^t], \quad \mathbb{E}[X^2 b + X a^t] \leq 172R. \quad (20)$$

Proof. Let $J = \mathbb{E}[b + X a^t] = b + a^t + t a^{t-1}$. Jensen gives $J^{t/2} \leq R$, and $a \leq \sqrt{J}$, $b \leq \sqrt{2J}$. On $X \leq 1/16$ the integrand is at most $(\sqrt{2} + 1/16)^t R \leq (71/48)^5 R < 8R$. This region has probability at most $1/16$, so contributes less than $R/2$. Its complement proves the first bound. For $f(u) = b + u a^t$, twice integrating by parts gives

$$\mathbb{E}[X^2 f(X)] = 2R + 4\mathbb{E}[X f'(X)] + \mathbb{E}[X^2 f''(X)].$$

The derivative bounds $|f'| \leq t a^{t-1}$ and $f'' \leq t(t-1) a^{t-2}$, together with

$$\mathbb{E}[X^2 a^{t-2} b + X a^{t-2}] \leq 2R + 2b^2 \mathbb{E}[X] + X a^{t-2} \leq 6R,$$

give $\mathbb{E}[X^2 f] \leq [2 + 4t\sqrt{6} + 6t(t-1)]R \leq (122 + 20\sqrt{6})R < 172R$. The zero-vector case is immediate. $\square$

Choose a unit eigenvector $\mathbf{e}$ for $\lambda_{\max}(\mathbf{Q})$ and let $\square = 1 - \lambda_{\max}(\mathbf{Q})$ and $\mathbf{u}_i = \text{tr}((\mathbf{I} - \mathbf{e}\mathbf{e}^T)\mathbf{Q}_i)$. Conditional application of Lemma 5.2, followed by Cauchy–Schwarz, yields

$$\mathbf{u}_i \leq 32 \sqrt{\square}. \quad (21)$$

Thus a nearly line-directed total stiffness forces each vertex-weighted stiffness to be nearly aligned with that line. No dimension or vertex-count constant occurs.

5.2 The two-case positivity proof

Set $k_{\square} = 3 + 2\sqrt{\square}$, $\square_0 = 1/4,000,000$, and $\delta = 2\square_0$. For $k \in [k_{\square}, k_{\square} + \delta]$, one has $24/5 < t = k - 1 < 5$. If $\square \geq \square_0$, then

$$\kappa(\mathbf{A}) \leq k - (k - 1)\square_0, \quad \kappa(\mathbf{B}_i) \leq k.$$

Let $g(k) = ((\sqrt{k} + 1)/(\sqrt{k} - 1))^2$. The turning angles have strictly positive dot product when $\kappa(\mathbf{A}) < g(k)$. Since $g(k_{\square}) = k_{\square}$ and $|g'(k)| \leq 1$ for $k \geq k_{\square}$,

$$k - g(k) \leq 2(k - k_{\square}) \leq 4\square_0 < (k - 1)\square_0.$$

This proves the first case.

If $\square \leq \square_0$, (21) gives $\mathbf{u}_i < 1/4$. In $\mathbb{R}^n \ni \mathbf{e}^{\square}$ write $\mathbf{Q}_i = \begin{bmatrix} q & \mathbf{b}^T \\ \mathbf{b} & \mathbf{C} \end{bmatrix}$, with $q = 1 - \mathbf{u}_i$, $\mathbf{C} \succeq 0$, $\text{tr } \mathbf{C} = \mathbf{u}_i$, and let $\mathbf{A}_0 = \text{diag}(k, \mathbf{I})$. Then

$$\text{Sym}(\mathbf{A}_0^{-1}\mathbf{B}_i) = \begin{bmatrix} (1 + tq)/k & \mathbf{v}^T \\ \mathbf{v} & \mathbf{D}_0 \end{bmatrix}, \quad \mathbf{v} = \frac{t(k+1)}{2k}\mathbf{b}, \quad \mathbf{D}_0 = \mathbf{I} + t\mathbf{C}.$$

Positive semidefiniteness gives $\mathbf{b}^T(\mathbf{I} + t\mathbf{C})^{-1}\mathbf{b} \leq q\mathbf{u}_i/(1 + t\mathbf{u}_i)$. Using $k \leq 6$, $t \geq 24/5$, and $\mathbf{u}_i \leq 1/4$, its Schur complement is at least

$$\frac{19}{24} - \frac{35}{12}\square^2 \frac{15}{176} = \frac{1689}{25344} > \frac{1}{16}.$$

Also $\square \mathbf{D}_0^{-1}\mathbf{v} \leq 4/3$ and $\mathbf{D}_0 \succeq \mathbf{I}$. Completing the square gives $\text{Sym}(\mathbf{A}_0^{-1}\mathbf{B}_i) \succeq \mathbf{I}/80$. Since $\square \mathbf{Q} - \mathbf{e}\mathbf{e}^T \succeq \square$, the inverse perturbation is at most $5\square$, while $\square \mathbf{B}_i \preceq 6$. Consequently

$$\text{Sym}(\mathbf{A}^{-1}\mathbf{B}_i) \succeq (1/80 - 30\square_0)\mathbf{I} \succeq \mathbf{I}/160.$$

This proves strict local positivity in the added interval. The kernel is continuously differentiable in the force variable for these $p > 2$ and $K \in \mathbb{T}\mathbf{I}$ at every nonconstant list, including collapsed lists. Integrating along a one-vertex path and joining any isolated total-coincidence point by continuity proves the finite law. The transfer theorem covers the remaining exponents down to p_- .

6 The all-shape triangle proof through sixteen

Theorem 6.1 (Triangle interval; exact integer certificates). *For every triangle in every finite ambient Euclidean dimension, with the fixed barycentric convention,*

$$\boxed{p_- \leq p \leq 16} \quad (22)$$

implies finite vertex attractivity. More quantitatively,

$$\begin{aligned} h \cdot \square U_p(\mathbf{P}_{i,h}) - U_p(\mathbf{P}) &\square \geq 10^{-10}\square h\square^2 & (10 \leq p \leq 16), \\ h \cdot \square U_{16}(\mathbf{P}_{i,h}) - U_{16}(\mathbf{P}) &\square \geq 10^{-9}\square h\square^2. & (23) \end{aligned}$$

These are conservative certified constants, not sharp minima.

We give the proof method and its finite certificate inventory; the complete stored covers and their independent replays are part of the proof [5, 6, 7, 8].

6.1 Normalization, root inclusion, and exact even-power integration

Normalize a longest edge to $\mathbf{A} = (-1, 0)$, $\mathbf{B} = (1, 0)$, and reflect or relabel so $\mathbf{C} = (\mathbf{a}, \mathbf{b})$ obeys

$$0 \leq \mathbf{a} \leq 1, \quad \mathbf{b} \geq 0, \quad (1 + \mathbf{a})^2 + \mathbf{b}^2 \leq 4. \quad (24)$$

The rectangle $\mathbf{D} = [0, 1] \times [0, 7/4]$ contains all such shapes, including flat limits and vertex collisions. An excluded box is accepted as exterior only if its lower endpoints satisfy the strict, exact inequality $(1 + \mathbf{a}_-)^2 + \mathbf{b}_-^2 > 4$. Equality is retained.

Let $\alpha = (\mathbf{p} - 2)/2$, $\mathbf{w} = \mathbf{x} - \mathbf{Y}_P$, and $r^2 = \mathbf{w} \cdot \mathbf{w}$. The kernels are

$$\mathbf{G} = \mathbb{E}[\mathbf{w}(r^2)^\alpha], \quad \mathbf{K} = \mathbb{E}[(r^2)^\alpha \mathbf{I} + 2\alpha(r^2)^{\alpha-1} \mathbf{w} \mathbf{w}^T], \quad \mathbf{L}_i = \mathbb{E}[\lambda_i((r^2)^\alpha \mathbf{I} + 2\alpha(r^2)^{\alpha-1} \mathbf{w} \mathbf{w}^T)].$$

The length-two base gives $\text{tr Cov}(\mathbf{Y}_P) \geq 1/6$. For $10 \leq \mathbf{p} \leq 16$, Jensen gives the unscaled global bound $\mathbf{K} \sqsubseteq 6^{-7} \mathbf{I}$, ensuring uniqueness and differentiability even at nonconstant flat configurations.

At a leaf, a fixed positive rational $\mathbf{R}^2$ may rescale all three kernels by $(\mathbf{R}^2)^{-\alpha}$. Writing $\rho = r^2/\mathbf{R}^2$ gives

$$\bar{\mathbf{G}} = \mathbb{E}[\mathbf{w} \rho^\alpha], \quad \bar{\mathbf{H}} = \rho^\alpha \mathbf{I} + (2\alpha/\mathbf{R}^2) \rho^{\alpha-1} \mathbf{w} \mathbf{w}^T, \quad \bar{\mathbf{K}} = \mathbb{E}[\bar{\mathbf{H}}], \quad \bar{\mathbf{L}}_i = \mathbb{E}[\lambda_i \bar{\mathbf{H}}].$$

The reference radius is constant in the spatial and vertex derivatives on that leaf; roots and $\mathbf{J}_i = \mathbf{K}^{-1} \mathbf{L}_i$ are unchanged. The continuous cover uses $\mathbf{R}^2 = \mathbf{k}/256$ with stored positive integers $\mathbf{k}$.

For a proposed rational point $\mathbf{x}_0$, an invertible rational preconditioner $\mathbf{M}$, and a rectangle $\mathbf{X} = \mathbf{x}_0 + [-\mathbf{r}_0, \mathbf{r}_0]$, the exact checker requires

$$-\mathbf{M} \bar{\mathbf{G}}(\mathbf{x}_0) + (\mathbf{I} - \mathbf{M} \bar{\mathbf{K}}(\mathbf{X}))(\mathbf{X} - \mathbf{x}_0) \sqsubseteq \text{int}(\mathbf{X} - \mathbf{x}_0). \quad (25)$$

Brouwer supplies a root for every parameter value in the shape/exponent box; strict convexity identifies it as the center. The same inclusion tightens the root enclosure before testing the response. No contraction assumption or numerical Newton convergence is used as a certificate.

The Duffy parametrization

$$\lambda = (1 - \mathbf{s}, \mathbf{s}(1 - \mathbf{t}), \mathbf{s}\mathbf{t}), \quad 0 \leq \mathbf{s}, \mathbf{t} \leq 1, \quad d\mathbb{P} = 2\mathbf{s} d\mathbf{s} d\mathbf{t} \quad (26)$$

turns the even-power kernels into polynomials. For $\mathbf{p} = 8, 10, 12$ the certificates use respectively 5, 6, and 7 Gaussian nodes in each coordinate, exact through degrees 9, 11, and 13. At $\mathbf{p} = 16$ the largest separate degree, including $2\mathbf{s}$, is 16; the 10×10 rule is exact through degree 19. The Legendre roots and weights are algebraic, enclosed rationally. Required moments are verified by exact polynomial quotient-algebra traces or the explicit quadratic field in the $\mathbf{p} = 8$ rule. Polynomial integration therefore has no truncation error before interval enclosure.

6.2 Two algebraic response tests

The first test encloses

$$\mathbf{Z}_i = \text{Sym}(\mathbf{M} \bar{\mathbf{K}} \bar{\mathbf{L}}_i \mathbf{M}^T) = (\mathbf{M} \bar{\mathbf{K}}) \text{Sym}(\mathbf{J}_i) (\mathbf{M} \bar{\mathbf{K}})^T.$$

Positive first diagonal and determinant certify positive definiteness. The quantitative lower bound is

$$\lambda_{\min}(\text{Sym } \mathbf{J}_i) \geq \frac{\det \mathbf{Z}_i}{\text{tr } \mathbf{Z}_i \sqcap \mathbf{M} \bar{\mathbf{K}} \sqcap_F^2}. \quad (27)$$

The second test is specific to two dimensions. Write

$$\bar{\mathbf{K}} = \begin{bmatrix} \mathbf{a} & \mathbf{c} \\ \mathbf{c} & \mathbf{b} \end{bmatrix}, \quad \bar{\mathbf{L}}_i = \begin{bmatrix} \mathbf{u} & \mathbf{w} \\ \mathbf{w} & \mathbf{v} \end{bmatrix}, \quad \mathbf{P}_i = \text{Sym}(\text{adj}(\bar{\mathbf{K}}) \bar{\mathbf{L}}_i).$$

Then $\mathbf{P}_i = (\det \bar{\mathbf{K}}) \text{Sym } \mathbf{J}_i$ and direct multiplication gives

$$\Delta_i := 4 \det \mathbf{P}_i = 4 \det \bar{\mathbf{K}} \det \bar{\mathbf{E}}_i - ((\mathbf{b} - \mathbf{a})\mathbf{w} + \mathbf{c}(\mathbf{u} - \mathbf{v}))^2. \quad (28)$$

Together with $(\mathbf{P}_i)_{11} = \mathbf{b}\mathbf{u} - \mathbf{c}\mathbf{w} > 0$, a strictly positive Δ_i proves the response positive. Its lower bound is

$$\lambda_{\min}(\text{Sym } \mathbf{J}_i) \geq \frac{\Delta_i}{4 \det \bar{\mathbf{K}} \text{tr } \mathbf{P}_i}. \quad (29)$$

The replay also verifies positive determinants of the stiffness matrices. This identity reduces interval overestimation from several matrix products and is the test used in the full continuous [10, 16] cover.

6.3 Every real exponent: the quadrature remainder

Endpoint positivity alone cannot prove an intervening interval. The continuous certificates explicitly partition shape **and** exponent space and enclose every real exponent on each closed leaf. The earlier cover uses $\alpha \in [j/32, (j+1)/32]$, $j = 77, \dots, 127$, giving $\mathbf{p} \in [109/16, 10]$. The later cover starts with $\mathbf{D} \times [4, 7]$, giving $\mathbf{p} \in [10, 16]$, and uses adaptive closed exponent intervals with endpoints in $\frac{1}{64}\mathbb{Z}$.

Here is the rigorous remainder used for the noninteger integrals. For $0 < \mathbf{l} \leq \mathbf{y} \leq \mathbf{u} < \mathbf{N}$, the shifted Chebyshev expansion on $[0, 1]$ is

$$\mathbf{z}^\gamma = \mathbf{c}_0 + \sum_{j \geq 1} \mathbf{c}_j \mathbf{T}_j(2\mathbf{z} - 1), \quad \mathbf{c}_j = 2\mathbf{c}_0 \sum_{r=0}^{j-1} \frac{\mathbf{y} - \mathbf{r}}{\mathbf{y} + \mathbf{r} + 1}.$$

If $\mathbf{m} = \lfloor \mathbf{l} \rfloor$, then $\mathbf{c}_0 \leq \frac{2^{\mathbf{m}}}{4^{\mathbf{m}}}$ and

$$\mathbf{C}_{N+1} = 2 \frac{2^{\mathbf{m}}}{4^{\mathbf{m}}} \sum_{r=0}^{\mathbf{N}} \frac{\max(|\mathbf{l} - \mathbf{r}|, |\mathbf{u} - \mathbf{r}|)}{\mathbf{l} + \mathbf{r} + 1}$$

bounds $|\mathbf{c}_{N+1}|$ uniformly. With $\mathbf{P}_N$ the truncated expansion,

$$|\mathbf{z}^\gamma - \mathbf{P}_N|_\infty \leq \mathbf{C}_{N+1} \frac{\mathbf{N} + \mathbf{u} + 1}{2\mathbf{l}}, \quad |\mathbf{z}(\mathbf{z}^\gamma - \mathbf{P}_N)|_\infty \leq \mathbf{C}_{N+1}. \quad (30)$$

To derive the coefficient recurrence, integrate the derivative of $\cos^{2\gamma+1}(\mathbf{s})\mathbf{e}^{i(2j+1)\mathbf{s}}$ on $[-\pi/2, \pi/2]$. The Fourier integrals obey $(\mathbf{y} + \mathbf{j} + 1)\mathbf{J}_{j+1} = (\mathbf{y} - \mathbf{j})\mathbf{J}_j$. Beyond $\mathbf{N}$, the signed coefficients alternate, decrease in magnitude, and their absolute sum telescopes to $|\mathbf{c}_{N+1}|(\mathbf{N} + \mathbf{y} + 1)/(2\mathbf{y})$. This proves the first estimate. For the second, let $\mathbf{z} = (1 + \cos \theta)/2$. Abel summation for the alternating tail, with the geometric-series bound, gives magnitude at most $|\mathbf{c}_{N+1}|/\sqrt{\mathbf{z}}$; multiplication by $\mathbf{z}$ proves the estimate, with $\mathbf{z} = 0$ handled by continuity. Integer powers terminate and obey the same bounds.

Use $\mathbf{N} = 8$ for $\mathbf{y} = \alpha$ to obtain a scalar error $\mathbf{e}_r$, and $\mathbf{N} = 7$ for $\mathbf{y} = \alpha - 1$ to obtain a weighted error $\mathbf{e}_l$. These choices satisfy $\mathbf{N} > \mathbf{u}$ both on the earlier strips and on $[4, 7]$. After including λ_i and $2\mathbf{s}$, every approximant has separate degree at most 18, so the degree-19 positive Gaussian rule integrates it exactly. If $\mathbf{B}$ bounds the squared physical distance from the enclosed center to every vertex, then convexity covers every material point. Put $\mathbf{Q} = \mathbf{B}/\mathbf{R}^2$. The true integral minus the quadrature of the rescaled nonpolynomial kernel has the following entrywise error bounds:

Entry	Absolute error bound
$\bar{\mathbf{G}}_j$	$2\sqrt{\bar{\mathbf{B}}} \mathbf{Q}^\alpha \mathbf{e}_r$
$\bar{\mathbf{K}}_{jj}$	$2\mathbf{Q}^\alpha (\mathbf{e}_r + 2\alpha \mathbf{e}_l)$
$\bar{\mathbf{K}}_{12}$	$2\alpha \mathbf{Q}^\alpha \mathbf{e}_l$
$\bar{\mathbf{E}}_{i,jj}$	$\frac{2}{3}\mathbf{Q}^\alpha (\mathbf{e}_r + 2\alpha \mathbf{e}_l)$
$\bar{\mathbf{E}}_{i,12}$	$\frac{2\alpha}{3}\mathbf{Q}^\alpha \mathbf{e}_l$

The factor two includes both integral and quadrature remainders; the off-diagonal uses $|w_1 w_2| \leq r^2/2$, and the vertex terms use $\mathbb{E}\lambda_i = 1/3$, exactly matched by the rule. All power factors and exponent maxima are bounded outward.

For a fixed nonnegative base, extremal powers over an exponent interval occur at endpoints. Dyadic endpoint powers are enclosed by integer powers and five or six successively enclosed square roots. Proof decisions use outward integer intervals on the 10^{-30} lattice, without floating transcendental evaluations.

6.4 Complete replay and finite-motion constants

Certificate	Positive boxes	Exterior exclusions	Total leaves
$p = 8$	19,789	0	19,789
$p = 10$	59,227	0	59,227
$p = 12$	30,906	477	31,383
$p = 16$	74,139	857	74,996
$p = 20$	548,619	2,103	550,722
$p = 21$	1,167,878	1,810	1,169,688
$[109/16, 10]$	415,426	12,851	428,277
$[10, 16]$	3,015,998	26,244	3,042,242

Table 2: Completed exact all-shape replays. The first two certificates test the whole containing rectangle without geometric exclusions. Every positive box checks the center and all three vertex responses.

The parent replay reconstructs every binary partition from its prescribed root, verifies all paths and boxes, and rejects duplicate, unused, or missing leaves. Exclusions use only the exact geometric inequality. Every other leaf must pass (25) and all three response tests. Numerical center guesses, inverse matrices, and radii are proposals only. Independent workers check disjoint assigned records; the parent verifies their input hashes, counts, and successful completion. A sampled replay is a diagnostic and cannot certify the theorem.

The exact uniform planar response bounds before simplification are

$$\begin{aligned} m_{10} &= \frac{84200067250756748723057}{5000000000000000000000000000000} > 10^{-7}, \\ m_{12} &= \frac{105219681913646386562653}{10^{30}} > 10^{-7}, \\ m_{16} &= \frac{28130703549804696210501}{20000000000000000000000000000000} > 10^{-7}, \\ m_{[109/16,10]} &= \frac{5491360738066882905017}{5000000000000000000000000000000} > 10^{-8}, \\ m_{[10,16]} &= \frac{10371718864833324830809}{10^{30}} > 10^{-8}. \end{aligned}$$

The $\mathbf{p} = 8$ replay establishes strict positivity without recording the same uniform quantitative response output.

In larger ambient space, $\mathbf{K}, \mathbf{L}_i$ split into the planar blocks and normal scalar blocks. The normal response equals

$$a_i = \frac{\mathbb{E}[\lambda_i r^{p-2}]}{\mathbb{E} r^{p-2}} > 0.$$

For $p \geq 2$, $K \leq (Er^{p-2})l$ and $L_i \leq (p-1)\mathbb{E}[\lambda_i r^{p-2}]l$, so $\|J_i\| \leq (p-1)a_i$. A planar lower bound m implies $a_i \geq m/(p-1)$. Integrating along any one-vertex path proves the finite law in every ambient dimension; continuity joins total coincidence. This yields the ambient constants 10^{-8} at $p = 10$, 10^{-9} at $p = 12$ and 16 , 10^{-9} throughout $[109/16, 10]$, and 10^{-10} throughout $[10, 16]$.

Finally $109/16 < p_+$, since $45/16 < 2^{\sqrt{2}}$. The transfer interval and the earlier continuous cover join to give $[p_-, 10]$; the later cover joins at the closed endpoint 10. This proves Theorem 6.1 for every real exponent, including 12, 14, 16. It does not infer a continuum of powers from isolated even-power results or from a monotonicity assumption.

7 The endpoints twenty and twenty-one, and the low-power investigation

7.1 Two further all-shape triangle theorems

Theorem 7.1 (Separate endpoint certificates; [9, 10]). *For every labelled triangle, every finite ambient Euclidean dimension, every vertex i , and every displacement h , the fixed barycentric law gives*

$$h \cdot \square U_{20}(P_{i,h}) - U_{20}(P) \square \geq 10^{-9} \square h \square^2, \quad (31)$$

$$h \cdot \square U_{21}(P_{i,h}) - U_{21}(P) \square \geq 10^{-11} \square h \square^2. \quad (32)$$

The proof uses the same compact shape domain, center inclusion, and complete cover audit as the preceding section. The completed covers are recorded in Table 2; their maximum depths are 24 and 26. Every accepted leaf checks all three vertex responses. Both replays use outward integer intervals on the 10^{-30} lattice and rational scale choices $R^2 = k/256$.

At $p = 20$, $\alpha = 9$. Force and vertex-weighted Hessian entries have separate degree at most 19 before the Duffy density is included, and at most 20 afterwards. A 12×12 Gaussian rule, whose degree-23 moments are verified exactly, therefore integrates the kernels without a quadrature truncation error. The congruence test (27) certifies the uniform planar lower bound

$$m_{20} = \frac{21997026509169687295293}{10^{30}} > 10^{-8}.$$

The fixed base supplies $K \square 6^{-9}I$, including nonconstant flat shapes.

At $p = 21$, $\alpha = 19/2$; exact polynomial integration is unavailable. Use the same degree-23 rule and the certified Chebyshev remainders (30), with degree 10 at $\gamma = 19/2$ and degree 9 at $\gamma = 17/2$. The approximants, including all weights, have separate degree at most 22. The rational remainder constants are

$$e_r = \frac{41327}{3158709829632}, \quad e_l = \frac{94809}{1667096854528}.$$

With $A = (B/R^2)^{19/2}$, the force error is at most $2^{\sqrt{B}} A e_r$. The diagonal and off-diagonal stiffness errors are, respectively, $2A(e_r + 19e_l)$ and $19Ae_l$; vertex-weighted errors are one third of these. These errors are included in every decision, not discarded as the shape boxes become smaller. Half-integer evaluation uses integer powers and an outward-enclosed square root. The commutator test (28) gives

$$m_{21} = \frac{150798289253456808433}{10^{29}} > 10^{-9}.$$

Here $K \square 6^{-19/2}I$. The normal-block estimate $a_i \geq m_p/(p-1)$ and integration along an arbitrary vertex segment give (31)–(32), including passages through degeneracy. This completes the proof from the accepted finite covers.

Independent implementation checks compare 864 polynomial entries for the even-power extension with Dirichlet monomial integration. For 21, they compare 324 entries with separate rational boundary integration, 864 neighboring even-power entries with exact moments, 800 dyadic power inequalities, and 108 entries at a symmetric flat shape with closed-form integrals. These checks supplement the complete replays; they do not replace coverage.

Corollary 7.2 (Unquantified open neighborhoods). Each of 20 and 21 lies in an open interval of all-shape attractive triangle exponents.

Proof. The normalized shape space is compact. Near either exponent, its nonconstant configurations have a uniform positive stiffness lower bound, and the force, root, and response matrices are jointly continuous in shape and exponent. The certified uniform strict margins persist in a sufficiently small common exponent neighborhood. The normal blocks and finite-path argument give the ambient finite law. $\square$

No explicit widths of these neighborhoods have been certified. They do not establish [16, 20] or [16, 21]. The largest explicitly proved continuous triangle interval is still $[p_-, 16]$.

7.2 What is known at $p = 65/64$

At this exponent $\alpha = -63/128$. The nonnegative-power remainder argument above cannot simply be reused for the singular interior kernel. The boundary representation (17) instead permits exact local certificates on proper triangles [9]. On a centered edge panel write the squared distance as $A(1+z)$ with $|z| \leq \beta \leq 1/8$. The binomial expansion of $(1+z)^\alpha$ through degree 35 has error bounded by

$$A^\alpha \left(\frac{\alpha}{36} \beta^{36} (1-\beta)^{\alpha-36} \right).$$

Reciprocals of exact integer-root enclosures bound A^α . For diameter at most two, the kernel has $H_p \square (p-1)2^{p-2} \square 1/128$ on the convex hull. Thus the force residual bounds the center error. A certified interior center ball stays separated from the edges; boundary derivative bounds then transfer the rational matrix enclosure to the true center.

For $A = (-1, 0)$ and $B = (1, 0)$ the following values of C have positive local response for every vertex and direction:

C	Center-error upper bound	Planar response lower bound
$(3/8, 5/8)$	10^{-13}	0.1522
$(3/4, 1/8)$	5×10^{-15}	0.1466
$(1/2, 1/256)$	4.7×10^{-11}	0.1438

The displayed bounds are conservatively rounded from rational enclosures. Each exact planar lower bound exceeds $1/10$; a conservative ambient local bound is $1/640$. These are local results at three specified shapes, not certificates for arbitrary finite displacements or for all shapes.

The flat reference $((-1, 0), (1, 0), (0, 0))$ has the exact responses

$$J_A = J_B = \frac{p}{2(p+1)} I = \frac{65}{258} I, \quad J_C = \frac{1}{p+1} I = \frac{64}{129} I \quad (p = 65/64).$$

A graded-boundary numerical search made 1,849 attempts: 1,675 passed its diagnostic thresholds, 169 remained diagnostically unresolved, and five solves failed. No resolved evaluation was negative. Its smallest value, approximately 0.2519379850, approaches $65/258$; arbitrarily thin shapes are not certified by this computation.

For contrast, a concrete **atomic** proper triangle at this same power is

$$A = (-3/5, -4/5), \quad B = (11/10, 4/5)(37/20)^{63/2}, \quad C = (-2^{-64}, 0).$$

Its center is exactly zero, because its three force vectors are $(3/5, 4/5)$, $(-11/10, -4/5)$, and $(1/2, 0)$. For $d = (1, 4)$ the exact stiffness enclosure proves

$$2^{63} d^T D_A M_{65/64} d < -75.$$

In particular the atomic rule fails locally and, by analyticity, for a sufficiently small finite motion. This witness specifies no numerical finite step size. It gives no negative conclusion for the hull rule. The all-shape hull question at $65/64$, and an extension below p_- , remain open.

8 Exact proper finite counterexamples

8.1 How a finite sign is certified

The negative certificates evaluate actual center displacements, not only approximate eigenvalues. If $\square^2 f_p \square \mu$ globally and $\mathfrak{q}$ is a rational approximation, then strong monotonicity gives

$$\|q - U_p\| \leq \|f_p(q)\|/\mu. \quad (33)$$

Thus two residual bounds $\mathbf{e}_0, \mathbf{e}_1$ imply the finite-response upper bound

$$h \cdot (U_p(P_{i,h}) - U_p(P)) \leq h \cdot (q_1 - q_0) + \lceil h \rceil (e_0 + e_1).$$

Exact polynomial moments, rational root enclosures, or explicit analytic integration remainders control every term. Properness at the endpoints and throughout the motion is checked separately.

Intrinsic d	p	Certified finite response	Certificate method
2	541/25	$< -10^{-10}$	Rational Taylor remainders
2	22	$< -10^{-8}$	Exact polynomial edge integrals
2	30	$< -10^{-7}$	Exact moments; see Section 9.2
3	20	$< -10^{-7}$	Symmetry and exact moments
5	20	$< -10^{-7}$	Rational polynomial cubature
31	20	$< -10^{-8}$	Proper lift and perturbation bound
12	18	$< -10^{-8}$	Regular-cluster exact moments
232	16	$< -10^{-13}$	Regular-cluster exact moments
302	16	$< -10^{-9}$	Regular-cluster exact moments

Table 3: Exact finite failures for proper uniform-solid simplices. A separate analytic triangle pair fails for every $\mathfrak{p} \geq 10^8$. Individual listed failures do not classify all larger exponents.

8.2 Acute triangles at 21.64 and 22

Take $\mathbf{A} = (-1, 0)$, $\mathbf{B} = (1, 0)$, $\mathbf{C}_0 = (359/500, 361/500)$, and $\mathbf{h} = (-1/500, 1/1000)$, with $\mathbf{C}_1 = \mathbf{C}_0 + \mathbf{h}$. At $\mathbf{p} = 541/25 = 21.64$,

$$h \cdot U_{541/25}(A, B, C_1) - U_{541/25}(A, B, C_0) \leq -\frac{117006294817184149611}{6103515625000000000000000000} < -10^{-10}. \quad (34)$$

Both endpoint triangles are acute and proper, and the path stays proper. The edge integrals have radial exponent $491/50$. Rational subdivisions write each squared distance as $\mathbf{a}(1+\mathbf{z})$ with $|\mathbf{z}| \leq 1/8$. Taylor expansion through degree 31 has the explicit remainder

$$\frac{491/50}{32} \beta^{32(1-\beta)^{\lceil 491/50 - 32 \rceil}}, \quad \beta \leq 1/8.$$

The factor $\mathbf{a}^{491/50}$ is enclosed by integer-power comparisons. A global Hessian bound $\boldsymbol{\mu} = 2^{-24}$ gives both center-error bounds through (33). At $\mathbf{p} = 22$, the same $\mathbf{h}$ with $\mathbf{C}_0 = (29/40, 29/40)$ gives a response below -10^{-8} ; all edge integrals are rational polynomials. These are independent exact finite certificates [3].

8.3 A proper tetrahedron and higher-dimensional examples at twenty

At $p = 20$, the tetrahedron

$$\mathbf{A} = (-1, 0, 0), \quad \mathbf{B} = (1, 0, 0), \quad \mathbf{C} = (-13/25, 14/25, 89/100), \quad \mathbf{D} = (-13/25, 14/25, -89/100)$$

fails when $\mathbf{A}$ moves by $\mathbf{h} = (-13/1000, 15/1000, 0)$. The certified finite response is at most approximately $-1.01296309777 \times 10^{-7}$ and strictly below -10^{-7} . Reflection forces the centers into $\mathbf{z} = 0$. Grouping the last two barycentric weights gives Dirichlet(1, 1, 2) and a conditional uniform variable $\mathbf{T} \in [-1, 1]$. All required moments are rational:

$$\mathbb{E}[\mathbf{u}^i \mathbf{s}^j] = \frac{6i!(j+1)!}{(i+j+3)!}, \quad \mathbb{E}T^{2r} = \frac{1}{2r+1}.$$

The covariance-trace lower bound $1/10$ yields $\mu = 10^{-9}$. Both center residuals certify errors below 10^{-25} .

An independent proper dimension-five example has vertices

$$\begin{aligned} &(-1, 0, 0, 0, 0), \quad (1, 0, 0, 0, 0), \quad (11/20, 71/100, \pm 87/100, 0, 0), \\ &(-1, 0, 0, 1/5, 0), \quad (-1, 0, 0, 0, 1/5). \end{aligned}$$

Move $\mathbf{B} = (1, 0, 0, 0, 0)$ by $(3/500, 1/125, 0, 0, 0)$. The finite response is at most approximately $-1.4005221008 \times 10^{-7}$, strictly below -10^{-7} . The edge determinant has absolute value 0.098832 before motion, and the path is proper. The verifier checks all 235 permutation classes of homogeneous degree-19 Dirichlet monomials at 5,005 rational cubature nodes. Symmetry and $\sum_i \lambda_i = 1$ establish every needed polynomial moment. Exact residuals after rational Newton refinement give center errors below 10^{-25} .

The earlier dimension-31 example starts with the projected Dirichlet(30, 1, 1) law at planar vertices $\mathbf{A} = (-1, 0)$, $\mathbf{B} = (1, 0)$, $\mathbf{C} = (49/100, 117/100)$. Move $\mathbf{C}$ by $(-3/1000, 1/1000)$. Replace the 30 copies near $\mathbf{A}$ by $\mathbf{A}_0 = \mathbf{A}$ and $\mathbf{A}_j = \mathbf{A} + 10^{-45} \mathbf{e}_{j+2}$, $j = 1, \dots, 29$, in $\mathbb{R}^{31}$, and append $\mathbf{B}, \mathbf{C}$. This is a proper simplex. A covariance-trace bound $5/1364$, a force-Lipschitz estimate, and the tiny but strictly positive spread bound the lift's center perturbation by 6.16×10^{-14} . The finite negative sign remains below -10^{-8} . No collapsed simplex is used as the final witness.

8.4 Regular clusters: proper failures at 18 and 16

In $\mathbf{E} = \mathbb{R}^3 \sqcup \{\mathbf{z} \in \mathbb{R}^m : \sum_j \mathbf{z}_j = 0\}$, of dimension $m+2$, take

$$\mathbf{A}_j = (-1, 0, 0, \varepsilon(\mathbf{e}_j - \mathbf{1}/m)), \quad \mathbf{B} = (1, 0, 0, 0), \quad \mathbf{C}, \mathbf{D} = (\mathbf{c}_x, \mathbf{b}, \pm \mathbf{c}, 0). \quad (35)$$

For $\mathbf{b}, \mathbf{c}, \varepsilon > 0$, the $m+3$ vertices form a proper simplex. The following parameter choices have exact finite certificates:

d	p	m	c_x	b	c	ε	Motion of $\mathbf{B}$ in xy
12	18	10	1/5	7/10	6/5	1/20	$(-3/500, -1/125)$
302	16	300	-3/20	77/100	37/25	1/100	$(27/5000, 21/2500)$

Their finite-response upper bounds are approximately $-6.2615694495 \times 10^{-8} < -10^{-8}$ and $-1.2007862551 \times 10^{-9} < -10^{-9}$, respectively. Permutation and reflection symmetries make the center planar. The grouped law is Dirichlet($\mathbf{m}, 1, 2$). Conditional on cluster mass, let $\theta \in \text{Dirichlet}(1, \dots, 1)$ and $\mathbf{W} = \sum_j \theta_j^2$. Its exact moments are

$$\mathbb{E}W^\ell = \frac{m!}{(m)_{2\ell}} [\mathbf{z}^\ell] \sum_{j=0}^{\ell} \frac{(2j)!}{j!} \mathbf{z}^j \binom{m}{j}, \quad (36)$$

where $(\mathbf{m})_r = \mathbf{m}(\mathbf{m}+1) \cdots (\mathbf{m}+r-1)$. Binomial expansion gives the moments of $\mathbf{W} - 1/\mathbf{m}$, the transverse squared norm before scaling. Combining these with cap moments $\mathbb{E}T^{2r} = 1/(2r+1)$ and Dirichlet moments reduces the force and Hessian to exact polynomials. For $\mathbf{n} = \mathbf{m} + 3$, along each motion a covariance-trace lower bound is

$$\frac{(2 - \mathbf{h}\mathbf{h}_1)^2}{2\mathbf{n}(\mathbf{n}+1)}.$$

Jensen and exact residuals enclose both centers with errors below 10^{-25} . The quotient tetrahedron stays proper: its determinant is proportional to $2\mathbf{c} \text{ cross}(\mathbf{A}, \mathbf{B}, \mathbf{C})$, positive at both endpoints and affine in the moved vertex. The transverse cluster remains independent with strictly positive spread. Neither certificate takes $\varepsilon \rightarrow 0$.

8.5 The improved proper witness in dimension 232

The same exact cluster-moment method gives a smaller failure dimension at $\mathbf{p} = 16$ [14]. In (35) take

$$\mathbf{m} = 230, \quad \varepsilon = \frac{1}{1000}, \quad \mathbf{c}_x = -\frac{59849}{500000}, \quad \mathbf{b} = \frac{95193}{125000}, \quad \mathbf{c} = \frac{732119}{500000},$$

and move $\mathbf{B}$ by

$$\mathbf{h} = \begin{bmatrix} -\frac{549903}{10^9} & -\frac{835228}{10^9} & 0 & 0 \end{bmatrix}.$$

There are 233 vertices in intrinsic dimension 232. Exact rational Newton refinement, polynomial force integration, and the global lower bound

$$\mathbf{K} \geq \frac{(2 - \mathbf{h}\mathbf{h}_1)^2}{2 \cdot 233 \cdot 234} \mathbf{I}$$

bound each center error by 2.6×10^{-33} . The accepted rational finite-response upper bound is approximately $-1.6385022523165632 \times 10^{-13}$, and is strictly less than -10^{-13} . The 229 cluster directions are independent and the quotient tetrahedron has positive determinant throughout the motion. The spread is the stated positive rational number; no limiting or floating-point sign argument is used in the final certificate. Consequently

$$\boxed{\mathbf{d}_{\min}(16) \leq 232.} \tag{37}$$

Here $\mathbf{d}_{\min}$ denotes the least intrinsic dimension admitting a proper failure. This is an upper bound, not a proof of minimality. The older dimension-302 certificate remains valid.

8.6 The infinite-power limit and a whole large-power failure range

Both models converge as $\mathbf{p} \rightarrow \infty$ to the minimum-enclosing-ball center of their support. For a proper simplex this is the ball enclosing its vertices. It need not be the circumcenter on an obtuse simplex. For a solid $\mathbf{d}$ -simplex of diameter $\mathbf{D}_0$, let $\mathbf{R}(\mathbf{x}) = \max_i \|\mathbf{x} - \mathbf{v}_i\|$ and let $\mathbf{O}$ be its enclosing-ball center with radius $\mathbf{R}$. A corner $(1 - \varepsilon)\mathbf{v}_i + \varepsilon\mathbf{S}$ has probability $\varepsilon^{\mathbf{d}}$, whence

$$[\mathbb{E}\|\mathbf{x} - \mathbf{Y}_P\|^{\mathbf{p}}]^{1/\mathbf{p}} \geq \varepsilon^{\mathbf{d}/\mathbf{p}}(\mathbf{R}(\mathbf{x}) - \varepsilon\mathbf{D}_0).$$

Optimality at $\mathbf{q} = \mathbf{U}_p$ gives

$$\mathbf{R}(\mathbf{q}) \leq \varepsilon^{-\mathbf{d}/\mathbf{p}}\mathbf{R} + \varepsilon\mathbf{D}_0, \quad \|\mathbf{q} - \mathbf{O}\|^2 \leq \mathbf{R}(\mathbf{q})^2 - \mathbf{R}^2. \tag{38}$$

The second inequality follows from strong convexity of the squared maximum-distance objective. First fixing ε , then sending it to zero, proves convergence.

A concrete pair with $A = (-1, 0)$, $B = (1, 0)$, $C_0 = (1/2, 1)$, $C_1 = (3/5, 99/100)$, and $h = (1/10, -1/100)$ has acute triangles and circumcenters $O_0 = (0, 1/8)$, $O_1 = (0, 3401/19800)$. Their response is $h \cdot (O_1 - O_0) = -463/990000$. With $\varepsilon = 10^{-8}$, (38) gives $\|U_p(P_j) - O_j\| < 0.00105$ for every real $p \geq 10^8$. Since $\|h\| < 0.101$, the solid finite response is below -0.00025 . Thus one fixed proper triangle pair fails throughout $[10^8, \infty)$. This coarse analytic threshold remains a theorem, although the isolated certificates at 21.64 and 22 are much closer to the observed branch.

9 Downward closure: proved reductions and a failed stronger assertion

Let $A(p)$ assert finite hull-power vertex attractivity for every triangle, with the fixed barycentric extension. The conjecture investigated here is

$$A(p_0) \Rightarrow A(p) \quad (4 < p < p_0). \quad (39)$$

It is not proved. If proved, it would join the endpoint 21 to the existing interval and establish $[p_-, 21]$. Neither a partially constructed continuation cover nor completed checks on only some of its segments establish that interval. We record the structural progress separately [11, 13, 12].

9.1 The frozen-angular-moment lemma

For fixed positive semidefinite 2×2 matrices R, S of trace one, put

$$d_R = \det R, \quad d_S = \det S, \quad c = (R_{22} - R_{11})S_{12} + R_{12}(S_{11} - S_{22}), \quad z(p) = \frac{(p-2)^2}{p-1}.$$

For $K = A(I + (p-2)R)$ and $L = B(I + (p-2)S)$, with $A, B > 0$, the commutator discriminant divided by $A^2 B^2 (p-1)^2$ equals

$$F_{R,S}(z) = 4 + 4(d_R + d_S)z + (4d_R d_S - c^2)z^2. \quad (40)$$

Because $\text{tr}(K^{-1}L) > 0$, the symmetric response is semidefinite exactly when $F_{R,S}(z) \geq 0$.

Lemma 9.1. If this response is semidefinite at $p_0 > 2$ with R, S fixed, it is positive definite at every $2 < p < p_0$.

Proof. The function $z(p)$ is strictly increasing for $p > 2$. If the quadratic coefficient in (40) is nonnegative, $F \geq 4$. Otherwise F is concave, and between zero and $z_0 = z(p_0)$ it lies above the chord joining $F(0) = 4$ to $F(z_0) \geq 0$; hence it is strictly positive. $\square$

In the actual hull problem, R and S are radial-weighted angular moments. They vary with both the moving center and the factor r^{p-2} . Thus this lemma does not compare the actual matrices at two exponents. Scalar log-convexity does not remove this dependence. Even Loewner monotonicity in the proposed direction fails: the symmetric flat triangle has outer vertex response $p/(2(p+1))$, increasing with p .

9.2 A fixed triangle and a fixed motion can regain positivity

Theorem 9.2 (Fixed-shape and finite-motion re-entry; [12]). For $A = (-1, 0)$, $B = (1, 0)$, $C = (13/20, 3/4)$, all local vertex responses are positive at 1, the C response is indefinite at 30, and all local vertex responses are positive for every sufficiently large p . For the single displacement $h = (-1/500, 1/1000)$ of C , the finite response is likewise positive at 1, negative at 30, and positive for every sufficiently large p .

Proof. The 21 assertion is Theorem 7.1. The exact polynomial certificate at 30 gives

$$4 \det K \det L_C - ((K L_C - L_C K)_{12})^2 \leq -\frac{696306873536785812603}{2 \cdot 10^{28}} < 0$$

and, independently, the finite-response upper bound $-1.4294882376917766 \times 10^{-7}$ approximately, strictly below -10^{-7} . Rational center residuals and $K \square 2^{-32}I$ control root error.

For the large-power assertion we prove the locally uniform C^1 expansion on the strict-obtuse domain $|a| < 1$, $b > 0$, $a^2 + b^2 < 1$:

$$U_p((-1, 0), (1, 0), (a, b)) = \frac{1}{p} \operatorname{atanh} a, \frac{b}{1-a^2} + o(1/p). \quad (41)$$

Put $x = (X/p, Z/p)$ and rescale the force as $(p^2 G_x, p^3 G_y)$. Only the two base corners contribute in the limit. At the first, use $\lambda_B = u/p$, $\lambda_C = v/p$; its radial factor tends to $\exp(X - 2u - (1+a)v)$ and its probability element is $2 du dv/p^2$. The other corner has factor $\exp(-X - 2u - (1-a)v)$. Away from them, the radius is uniformly less than one and contributions decay exponentially. In the corners, the integrands and first parameter derivatives are bounded by a polynomial times $Ce^{-c(u+v)}$. Dominated convergence gives the limiting force

$$\frac{e^X}{1+a} - \frac{e^{-X}}{1-a}, \quad \frac{e^X}{1+a} Z - \frac{b}{1+a} + \frac{e^{-X}}{1-a} Z - \frac{b}{1-a}.$$

Its unique zero is $(\operatorname{atanh} a, b/(1-a^2))$, with nonsingular triangular derivative. Stability of this zero in C^1 and strict convexity prove (41). Smooth normalization of moving base endpoints also gives $J_A, J_B \rightarrow 1/2$. Writing $D = 1 - a^2$, the apex response satisfies

$$pJ_C \rightarrow \frac{D^{-1}}{2abD^{-2}} \frac{0}{D^{-1}}, \quad \lambda_{\min} \operatorname{Sym}(pJ_C) \rightarrow \frac{D - |ab|}{D^2}.$$

For the stated triangle this last value is $1600/5929 > 0$. Along $C + th$, $0 \leq t \leq 1$, strict obtuseness persists and $1 - a_t^2 - a_t b_t \geq 9/100$. If $f(a, b)$ is the vector in (41), then $\operatorname{Sym} D f \square (9/100)I$ on this segment. Consequently

$$\lim_{p \rightarrow \infty} p h \cdot U_p(A, B, C + h) - U_p(A, B, C) = h \cdot (f(C + h) - f(C)) \geq \frac{9}{20,000,000} > 0.$$

□

No particular finite exponent of return is certified by the asymptotic proof. Numerical edge quadrature suggests a return between 1500 and 2000. The theorem rules out fixed-shape no-re-entry, but not (39): an all-shape assertion concerns the intersection over every triangle. A bad triangle may change with the exponent. The same asymptotic formula supplies eventual apex failure whenever $|ab| > 1 - a^2$ within the strict-obtuse domain.

9.3 An exact disk criterion for proper triangles

Let $x = U_p(T)$, $y = Y - x$, $r = |y|$, $t = p - 2 > 0$, and write the barycentric functions as $\lambda_i(Y) = a_i + g_i \cdot y$, with $a_i = \lambda_i(x) > 0$. Stationarity immediately gives

$$\mathbb{E}[\lambda_i r^{p-2}] = a_i \mathbb{E} r^{p-2}. \quad (42)$$

Define the symmetric third-order tensor $C = \mathbb{E}[r^{p-4} y^{\square 3}]$. Its trace is zero by stationarity, so write $C_{111} = c_1$, $C_{112} = c_2$, $C_{122} = -c_1$, $C_{222} = -c_2$. Set

$$K = kI + Q, \quad Q = \begin{pmatrix} q_1 & q_2 \\ q_2 & -q_1 \end{pmatrix}, \quad D = \det K > 0, \quad d = (q_1 c_1 + q_2 c_2, q_1 c_2 - q_2 c_1).$$

Here $\mathbf{C}(\mathbf{g})$ denotes contraction with a vector $\mathbf{g}$, namely

$$\mathbf{C}(\mathbf{g}) = \begin{pmatrix} c_1 g_1 + c_2 g_2 & c_2 g_1 - c_1 g_2 \\ c_2 g_1 - c_1 g_2 & -c_1 g_1 - c_2 g_2 \end{pmatrix}.$$

Equation (42) yields $\mathbf{L}_i = \mathbf{a}_i \mathbf{K} + \mathbf{tC}(\mathbf{g}_i)$. Since $\text{Sym}(\mathbf{QC}(\mathbf{g})) = (\mathbf{d} \cdot \mathbf{g}) \mathbf{I}$ and $\mathbf{C}(\mathbf{g})^2 = (c_1^2 + c_2^2) |\mathbf{g}|^2 \mathbf{I}$, multiplication by $\mathbf{K}^{-1}$ gives

$$\lambda_{\min} \text{Sym } \mathbf{J}_i = \mathbf{a}_i - \frac{\mathbf{t} \mathbf{d} \cdot \mathbf{g}_i}{D} - \frac{\mathbf{t} \mathbf{k} \sqrt{c_1^2 + c_2^2}}{D} |\mathbf{g}_i|. \quad (43)$$

Theorem 9.3 (Moment disk; [13]). For a proper triangle and $p > 2$, define

$$D_p(\mathbf{T}) = \overline{\mathbf{B}} \mathbf{x} - \frac{\mathbf{t} \mathbf{d}}{D}, \frac{\mathbf{t} \mathbf{k} \sqrt{c_1^2 + c_2^2}}{D}.$$

All three symmetric local responses are positive semidefinite exactly when $D_p(\mathbf{T}) \in \mathbf{T}$, and positive definite exactly when $D_p(\mathbf{T})$ lies in the interior of $\mathbf{T}$.

Proof. The minimum of an affine barycentric function on a disk of center $\mathbf{z}$ and radius ρ is $\lambda_i(\mathbf{z}) - \rho |\mathbf{g}_i|$. This is precisely (43). The three nonnegative barycentric half-planes intersect in $\mathbf{T}$. Normal ambient responses equal $\mathbf{a}_i > 0$ by (42). $\square$

This is a geometric reformulation at each exponent, not yet a comparison of the disks at different exponents.

9.4 Equal edge moments and elimination of lower moments

For an interior point $\mathbf{x}$, put

$$I_{jk}(\mathbf{x}, p) = \int_0^1 |(1-s)\mathbf{v}_j + s\mathbf{v}_k - \mathbf{x}|^p ds.$$

If N_{jk} is edge length times the outward unit normal, divergence gives

$$\mathbf{G}_p(\mathbf{x}) = -\frac{1}{p|\mathbf{T}|} \sum N_{jk} I_{jk}.$$

The three vectors N_{jk} have rank two and sum to zero. Hence

$$\mathbf{x} = \mathbf{U}_p(\mathbf{T}) \iff I_{12} = I_{23} = I_{31}. \quad (44)$$

At the center let I be this common value and write one edge's squared radius as $q(s) = \alpha + \beta s + \gamma s^2$, with endpoint powers $R_j = q(0)^{p/2}$, $R_k = q(1)^{p/2}$. For the lower moments $\mathbf{A} = \int_0^1 q^{(p-2)/2}$, $\mathbf{B} = \int_0^1 s q^{(p-2)/2}$, $\mathbf{C}_e = \int_0^1 s^2 q^{(p-2)/2}$, integration of two elementary derivatives gives

$$\begin{aligned} \beta \mathbf{A} + 2\gamma \mathbf{B} &= 2(R_k - R_j)/p, \\ \alpha \mathbf{A} + \beta \mathbf{B} + \gamma \mathbf{C}_e &= I, \\ \beta \mathbf{B} + 2\gamma \mathbf{C}_e &= 2(R_k - I)/p. \end{aligned}$$

The determinant is $-\gamma(4\alpha\gamma - \beta^2) < 0$, because the center is off the edge line. Thus these moments, and hence all boundary stiffness entries, reduce to linear combinations of I, R_j, R_k . These remaining quantities still depend on the exponent.

9.5 Unique centered projective families in every dimension

Fix a proper d -simplex T containing zero, $d \geq 2$, and define

$$\Omega_T = \{b : 1 + b \cdot v_i > 0 \text{ for every } i\}, \quad F_b(z) = \frac{z}{1 + b \cdot z}, \quad T_b = F_b(T) = (T^\circ + b)^\circ.$$

These maps preserve rays and line segments; their Jacobian is $(1 + b \cdot z)^{-(d+1)}$.

Theorem 9.4 (Centered comparison family; [13]). **For every $p > 1$ there is a unique $b_p \in \Omega_T$ with $U_p(T_{b_p}) = 0$. The map $p \mapsto b_p$ is smooth. Up to common scale, T_{b_p} is the unique simplex with the prescribed vertex rays and power center zero.**

Proof. Consider

$$\Phi_{p,d}(b) = \int_T \frac{|z|^{p-2}}{(1 + b \cdot z)^{p+d-1}} dz = \int_{T_b} |y|^{p-2} dy.$$

Differentiation gives

$$\nabla \Phi_{p,d} = -(p + d - 1) \int_{T_b} |y|^{p-2} y dy,$$

and a positive-definite Hessian, equal to $(p + d - 1)(p + d) \int_T |z|^{p-2} z z^T / (1 + b \cdot z)^{p+d+1} dz$. The domain Ω_T is bounded. At a boundary-approaching vertex let its denominator be ε . On a corner of volume proportional to ε^d , the denominator is $O(\varepsilon)$ and $|z|$ is bounded away from zero. Thus $\Phi_{p,d} \geq c\varepsilon^{1-p} \rightarrow \infty$. Strict convexity and coercivity give a unique minimizer; its stationarity is the center equation. The implicit-function theorem gives smoothness. All derivatives are locally integrable at zero for $d \geq 2$, $p > 1$.

For the final assertion, let $a_i > 0$ be the origin's barycentrics in T . Given new positive vertex factors t_i , put $s = (\sum_i a_i/t_i)^{-1}$. The prescribed affine values $1 + b \cdot v_i = s/t_i$ have barycentric average one and determine a unique $b \in \Omega_T$. Then $t_i v_i = s F_b(v_i)$. $\square$

Differentiating the centered force equation yields the useful identity

$$b_p = \frac{1}{p + d} \int_{T_{b_p}} r^{p-2} y y^T dy - \frac{1}{p + d} \int_{T_{b_p}} r^{p-2} y \log r dy. \quad (45)$$

There is no immediate sign implication.

This construction has an established convex-geometric interpretation. For $Q = T^\circ$, $q = p + d - 2$, and $\alpha = 1/(p + d - 1)$, polar integration gives

$$q \Phi_{p,d}(b) = \int_{S^{d-1}} h_{Q+b}(u)^{-q} du.$$

Thus $-b_p$ is the α -entropy point of Q , in the terminology of Andrews, Guan, and Ni [18], Section 2, particularly Proposition 2.3. The proof above is self-contained. Their scalar entropy monotonicity does not give the vertex-response matrix inequality required here.

9.6 Eventual failure along every nonregular ray family

Theorem 9.5 (Large-power projective limit; [13]). **The family T_{b_p} converges to the unique equal-radius simplex with the prescribed rays, up to the fixed scale convention. If that simplex is nonregular, some local vertex response is negative at every sufficiently large p . The regular family has positive local responses at every $p > 1$.**

Proof. Let a_i be the origin's barycentrics of the reference simplex, $r_i = |v_i|$, and $R_\square = \sum_i a_i r_i$. The equations $1 + b_\square \cdot v_i = r_i/R_\square$ determine $b_\square \in \Omega_T$. For any b , if $R = \max_i r_i/(1 + b \cdot v_i)$, then

$$1 = \sum_i a_i (1 + b \cdot v_i) \geq R^{-1} \sum_i a_i r_i = R_\square/R.$$

Equality forces $\mathbf{b} = \mathbf{b}_\square$. Polar integration identifies minimization of $\Phi_{p,d}$ with minimization of the L^q norm of the radial function, $q = p + d - 2$. These minimizers stay in a compact subset of Ω_T : if a largest vertex radius is M , an adjacent angular cap of measure at least $\mathbf{c}M^{-(d-1)}$ has radius at least M/C . Hence its L^q norm is at least $\mathbf{c}^{1/q}M^{1-(d-1)/q}/C$, whereas the competitor $\mathbf{b}_\square$ has norm at most $|\mathbf{S}^{d-1}|^{1/q}R_\square$. For $p \geq 2$ the exponent of M is at least $1/d$. All radii are therefore bounded; denominators stay away from zero. Uniform convergence of the radial L^q norms to their maxima on this compact set implies $\mathbf{b}_p \rightarrow \mathbf{b}_\square$.

The limiting simplex has an interior circumcenter at zero. On a compact family of such simplices a corner Laplace expansion gives

$$\mathbf{U}_p(\mathbf{T}) = \mathbf{O}(\mathbf{T}) + \mathbf{O}(1/p) \quad \text{in } \mathbf{C}^1, \quad (46)$$

where $\mathbf{O}$ is the circumcenter. To see this, center the circumcircle of radius R at zero and put $\mathbf{x} = \mathbf{u}/p$. The limiting corner force, after a common positive normalization, is

$$\square_i \kappa_i \mathbf{e}^{-\mathbf{v}_i \cdot \mathbf{u}/R^2} \mathbf{v}_i, \quad \kappa_i = \square \frac{d!}{\prod_{j \neq i} (1 - \mathbf{v}_i \cdot \mathbf{v}_j / R^2)} > 0.$$

Its positive coefficients must be proportional to the circumcenter's positive barycentrics. Their ratios determine $\mathbf{u}$ uniquely, and its derivative is a nonsingular multiple of $\square_i \mathbf{a}_i \mathbf{v}_i \mathbf{v}_i^T$. Corner integrands and first shape derivatives admit uniform polynomial times exponential bounds; the complement decays exponentially. Expanding one further order and applying the implicit-function theorem proves (46), including vertex derivatives.

Let $\mathbf{n}_i$ point normally from the opposite facet toward $\mathbf{v}_i$, and let $h_i > 0$ be the altitude. At a circumcenter normalized to zero,

$$\mathbf{D}_{\mathbf{v}_i} \mathbf{O} = \frac{\mathbf{n}_i \mathbf{v}_i^T}{h_i}, \quad \lambda_{\min} \text{Sym } \mathbf{D}_{\mathbf{v}_i} \mathbf{O} = \frac{\mathbf{n}_i \cdot \mathbf{v}_i - |\mathbf{v}_i|}{2h_i} \leq 0.$$

Equality holds only when $\mathbf{v}_i$ is parallel to $\mathbf{n}_i$. If this holds for every vertex, all off-diagonal Gram entries are equal; equal radii then force all edge lengths equal. Otherwise some response is strictly negative, and (46) transfers that sign to $\mathbf{T}_{b_p}$ for all sufficiently large p . In the regular case symmetry makes $\mathbf{K}$ scalar and every $\mathbf{L}_i$ positive definite, so the local responses are positive. $\square$

The threshold may depend on the family and dimension. This theorem does not rule out intermediate re-entry and supplies no dimension-independent failure threshold.

9.7 The remaining crossing inequality

For the centered triangle family, take unnormalized integrals, put $\mathbf{W} = \mathbf{r}^{p-2}$, $\mathbf{e} = \mathbf{y}/r$, $\mathbf{E} = \mathbf{e}\mathbf{e}^T$, $\mathbf{H} = \mathbf{I} + (p-2)\mathbf{E}$, and $\beta = \mathbf{b}_p^\square$. The composition identity $\mathbf{F}_c \circ \mathbf{F}_b = \mathbf{F}_{b+c}$ gives

$$\dot{\mathbf{v}}_i = -(\beta \cdot \mathbf{v}_i)\mathbf{v}_i, \quad \dot{\mathbf{a}}_i = (\beta \cdot \mathbf{v}_i)\mathbf{a}_i, \quad \dot{\mathbf{g}}_i = (\beta \cdot \mathbf{v}_i)\mathbf{g}_i - \mathbf{a}_i\beta.$$

Changing variables under the projective map and differentiating yields

$$\begin{aligned} \dot{\mathbf{K}} &= \square \mathbf{W} \{ \mathbf{H} [\log r - (p+1)\beta \cdot \mathbf{y}] + \mathbf{E} \} \mathbf{d}\mathbf{y}, \\ \dot{\mathbf{L}}_i &= \square \lambda_i \mathbf{W} \{ \mathbf{H} [\log r - (p+2)\beta \cdot \mathbf{y} + \beta \cdot \mathbf{v}_i] + \mathbf{E} \} \mathbf{d}\mathbf{y}, \\ \dot{\mathbf{J}}_i &= \mathbf{K}^{-1}(\dot{\mathbf{L}}_i - \dot{\mathbf{K}} \mathbf{J}_i). \end{aligned} \quad (47)$$

Let $\mathbf{f}_i = \lambda_{\min}(\text{Sym } \mathbf{J}_i)/\mathbf{a}_i$. At $\mathbf{f}_i = 0$ the zero eigenvalue is simple, since $\text{tr } \mathbf{J}_i > 0$. For its unit eigenvector $\mathbf{h}$,

$$\mathbf{f}_i^\square = \frac{1}{\mathbf{a}_i} \mathbf{h}^T \mathbf{K}^{-1}(\dot{\mathbf{L}}_i - \dot{\mathbf{K}} \mathbf{J}_i) \mathbf{h}. \quad (48)$$

Proposition 9.6 (Conditional crossing criterion). If (48) is strictly negative at every zero $f_i = 0$ for every proper centered triangle and every $p > 4$, then (39) holds, with strict local positivity at every lower exponent on proper triangles.

Proof. A function f_i that is negative cannot first return to zero with strictly negative derivative; a starting zero also immediately enters the negative side. Thus a nonpositive lower response would generate a negative upper response on its centered fixed-ray family, contradicting $A(p_0)$. Collinear nonconstant configurations have $J_i = (\mathbb{E}[\lambda_i r^{p-2}] / \mathbb{E} r^{p-2}) I \leq 0$, because all kernel matrices share the same axial form. Integrating local positivity and joining total coincidence by continuity gives the finite law. $\square$

The hypothesis is unproved. It is weaker than requiring f_i to decrease everywhere, which numerical diagnostics already disfavor away from zero. The expanded numerical probe attempted 1,500 families; 1,404 completed sampled continuation through 1000, and 96 had unresolved solver failures. Among 23,178 evaluated family/exponent cases it located 3,375 zero crossings, all with negative slope; the largest was approximately -4.55163×10^{-6} . Recalculation at quadrature order 384 changed slopes by at most 1.18×10^{-12} . These are diagnostics, not a universal crossing certificate or an exclusion of missed roots. Independent algebra checks cover 100 exact disk identities; volume/boundary and derivative comparisons further check the formulas.

10 Systematic numerical searches

All numbers in this section are numerical search outputs, not rigorous global minima. Positive samples do not exclude a missed branch, and connecting numerical branch values does not prove an exponent interval.

10.1 Triangles

The normalized longest-edge representation is $C = (a, f \sqrt{4 - (1 + a)^2})$, $0 \leq a, f \leq 1$. A deterministic grid with 3,969 pairs per exponent, combining linear spacing with logarithmic approaches down to 10^{-8} at every edge, made 47,628 coverage evaluations at 12 powers. Six separated low-valued seeds per power were refined locally, testing every vertex response. There were no solver or diagnostic failures. Selected smallest found eigenvalues were:

p	Smallest found symmetric-response eigenvalue
6.828428	0.12773956
7	0.12500000
8	0.11111111
10	0.08986347
12	0.07177646
16	0.03966577
18	0.02499972
20	0.01100354
21.5	0.00088513
21.62	0.00008855
21.633	0.00000237

The symmetric flat triangle $(-1, 0), (1, 0), (0, 0)$ has middle-vertex response $1/(p+1)$ for $p > 2$: its material density is $1 - |y|$ and its conditional mean middle barycentric weight is $(1 - |y|)/2$. The corresponding weighted moment ratio is exactly $1/(p+1)$, explaining the observed boundary values at 7 and 8.

Local continuation recovered an upper failure branch near $\mathbf{C} = (0.7178, 0.7218)$, with an apparent sign change between 21.63333984 and 21.63335938. The systematic search found no negative response at the tested powers through 21.633. The rigorous conclusion about positive intervals is still only the range through 16. The exact 21.64 counterexample confirms failure nearby without proving the observed branch to be the first or sharp one.

10.2 Tetrahedra

With $\mathbf{A} = (-1, 0, 0)$, $\mathbf{B} = (1, 0, 0)$, $\mathbf{C} = (\mathbf{c}_x, \mathbf{c}_y, 0)$, $\mathbf{D} = (\mathbf{d}_x, \mathbf{d}_y, \mathbf{d}_z)$, the longest-edge search has five shape parameters. For each of eight powers it used 8,192 Halton coverage points, 2,048 boundary approaches, and eight retained branch seeds. Twelve separated low-valued seeds and all eight branch seeds were locally refined. The total was 81,984 coverage and 48,830 local evaluations, plus 24 final quadrature-order comparisons.

p	Refined smallest found response, checked at order 22
8	0.09543851769
12	0.06454010517
16	0.03305490076
18	0.01628787790
19	0.007934932914
19.5	0.003773559529
19.9	0.0004532142588
19.95	0.00003875952400

Orders 14, 18, 22 checked each winning shape. At 8, 12 the winners approach flat limits; higher powers recover the proper upper branch. Earlier continuation gives positive response at 19.95 and negative response at 19.96, suggesting onset near 19.9547. This is not a sharp threshold or an all-shape positivity proof. The exact tetrahedral failure at 20 is independent of that numerical interpretation. In particular, the initial finite search at 20 missed a negative branch later recovered by continuation, illustrating the limits of sampling.

10.3 Proper higher-dimensional families

Full-response multistart searches in dimensions 4, 5, 6 examined two regular cap groups, tetrahedral cluster lifts, and general affine simplices with strong anisotropy, at $\mathbf{p} = 8, 12, 16, 18, 20$. There were 6,855 coverage and 50,790 local evaluations; 60 winners were checked by positive tensor Gaussian integration. Signed polynomial cubature accelerated the search and was independently compared with positive integration and finite differences.

The regular-cluster family (35) was scanned in dimensions 4, 5, 6, 8, 12, 32, 102, using 291 coverage shapes per power/dimension case and three refinements, for 10,185 coverage evaluations. Spread was constrained to be at least 0.05, so these were proper shapes. This reduction examines the response of $\mathbf{B}$; it does not certify all individual cluster vertices.

d	Smallest searched $\mathbf{B}$ response at $\mathbf{p} = 18$	At $\mathbf{p} = 20$
4	0.01205014	-0.00389365
5	0.00895757	-0.00635293
6	0.00657577	-0.00816523
8	0.00316506	-0.01058483
12	-0.00066927	-0.01284543
32	-0.00437371	-0.01223480
102	-0.00283691	-0.00635842

All scanned $\mathbf{B}$ responses at 8, 12, 16 in this spread-constrained family were positive. The targeted $\mathbf{m} = 300$ smaller-spread follow-up found the proper $\mathbf{p} = 16$ failure subsequently certified in dimension 302. Additional earlier diagnostics used 14 coalescing-vertex Dirichlet patterns (48,384 initial evaluations), 604 tetrahedral samples with boundary and anisotropic approaches, and regular-fan families in dimensions 3, 4, 6, 11, 31, 101. Projected limits in those studies are explicitly distinguished from the proper finite witnesses above.

11 What is proved about dimension dependence

Let $\mathbf{A}_d$ denote the set of exponents for which the uniform-solid rule on proper $\mathbf{d}$ -simplices has the global finite law, with the fixed barycentric extension used for its proof. Let $\mathbf{A}_{\text{all}} = \bigcap_{d \geq 1} \mathbf{A}_d$. The results give

$$[\mathbf{p}_-, \mathbf{p}_+ + \delta] \subseteq \mathbf{A}_{\text{all}}, \quad (1, \infty) = \mathbf{A}_1, \quad [\mathbf{p}_-, 16] \subseteq \{20, 21\} \subseteq \mathbf{A}_2,$$

and

$$21.64, 22 \notin \mathbf{A}_2, \quad 20 \notin \mathbf{A}_3, \quad 18 \notin \mathbf{A}_{12}, \quad 16 \notin \mathbf{A}_{232}, \quad 16 \notin \mathbf{A}_{302}.$$

In particular $\mathbf{A}_2 \not\subseteq \mathbf{A}_{232}$ and $16 \notin \mathbf{A}_{\text{all}}$. Strict finite failures persist in some exponent neighborhood by continuity. These facts do not prove that the sets are connected, nested in $\mathbf{d}$, or described by a common reciprocal or dimension-specific endpoint formula. If one first proved the common set to be a single attractive interval, its upper endpoint would have to lie below 16; connectedness is presently an additional hypothesis.

Ambient embedding dimension plays a different role. An embedded triangle has the same planar response and positive normal scalar responses, so Theorems 6.1 and 7.1 are independent of $\mathbf{D}$. Adding vertices to form a higher-dimensional proper simplex changes the barycentric probability law, including projected group multiplicities. It is not an embedding of the same triangle measure.

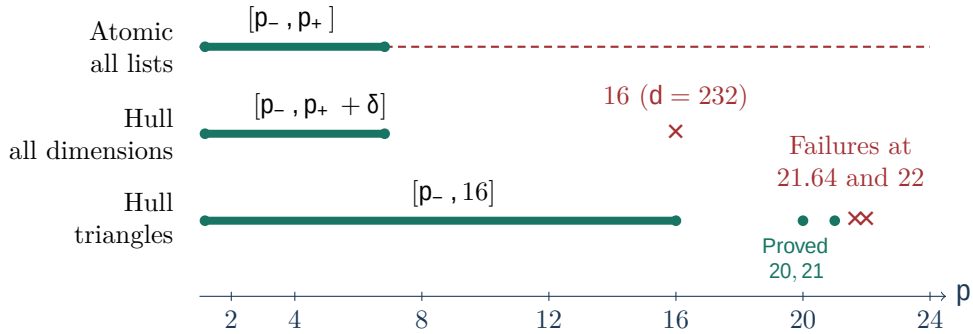


Figure 1: Proved intervals (solid green), isolated proved endpoints (green dots), and certified obstructions. Atomic dashed portions are excluded by the exact classification. Hull failure marks are individual exponents, not excluded half-lines. The tiny common extension δ is not visually resolved at this scale. Blank hull portions carry no positive or negative claim.

11.1 Minimal failure dimensions and upper failure powers

For proper simplices define

$$\mathbf{d}_{\min}(\mathbf{p}) = \min\{\mathbf{d} \geq 1 : \mathbf{p} \notin \mathbf{A}_d\}, \quad \mathbf{p}_{\min}(\mathbf{d}) = \inf\{\mathbf{p} > 4 : \mathbf{p} \notin \mathbf{A}_d\},$$

with value $+\infty$ if the relevant set is empty. The second definition concerns only the upper branch; it makes no connectedness assumption. An explicit failure supplies an upper bound. A search that finds no negative example supplies no global lower bound.

For each integer $p = 7, 8, 9, 10, 11, 12, 13, 14, 15$, no failure witness was found in the dimension investigation [14]; no finite upper bound on $d_{\min}(p)$ has been established. For $p = 16$, the exact result is (37). The triangle theorem gives $d_{\min}(p) \geq 3$ when finite throughout these powers. In particular, $d_{\min}(7)$ is unresolved: selected-family searches reaching dimension about one million do not establish $d_{\min}(7) > 10^6$ or $d_{\min}(7) = \infty$.

The main numerical construction groups $m = d - 2$ copies of $A = (0, 0, 0)$, one $B = (1, 0, 0)$, and two caps $(u, v, \pm c)$, $v, c > 0$. Its projected law is Dirichlet($m, 1, 2$). Every individual vertex type was tested, including the mixed planar/transverse block of a cap response. A small regular spread of the A group in an orthogonal $(m - 1)$ -space gives a proper d -simplex. At $p > 4$, strict negative responses persist under sufficiently small spreads by continuity. The numerical boundary uses the collapsed limit to estimate onset; specified proper finite witnesses require the additional error and spread control supplied by the exact certificates.

For noninteger powers the conditional cap integral is evaluated using

$$\mathbb{E}[(A + ZT^2)^q] = (A + Z)^q {}_2F_1\left(-q, 1; \frac{3}{2}; \frac{Z}{A + Z}\right), \quad T \sim \text{Unif}[-1, 1].$$

The second- T -moment replaces $3/2$ by $5/2$ and includes a factor $1/3$. Positive Gaussian beta quadrature handles the remaining two integrals. Orders 24, 36, and 52 differed by less than 9.4×10^{-10} in the scaled response minima at the reported failing powers. These are numerical convergence checks, not interval enclosures.

Intrinsic d	Estimated onset	Numerically checked failing p
2	21.6334	21.7
3	19.9547	20.0
4	19.5061	19.6
5	19.1657	19.2
6	18.8891	18.9
8	18.4589	18.5
10	18.1363	18.2
12	17.8844	17.9
16	17.5160	17.6
20	17.2597	17.3
30	16.8664	16.9
50	16.5007	16.6
75	16.2980	16.3
100	16.1911	16.2
150	16.0804	16.1
200	16.0236	16.1
232	15.9997	16.0
250	15.9890	16.0
300	15.9657	16.0
500	15.9187	16.0
1,000	15.8829	15.9
10,000	15.8504	15.9
100,000	15.8472	15.9
1,000,000	15.8469	15.9

Table 4: Numerical family onset and an actually checked negative power, rounded upward to one decimal. These are not certified global values of $p_{\min}(d)$. Exact results are identified in the text.

Dimension one has no failing power. Exact certificates already give $p_{\min}(2) \leq 21.64$, $p_{\min}(3) \leq 20$, $p_{\min}(12) \leq 18$, and $p_{\min}(232) \leq 16$. For example, 17.9 at $d = 12$ is a numerical improvement on the exact witness at 18; it is not presented as a new exact certificate. Similarly, 16.5 at $d = 50$ would round **below** the estimated onset and is deliberately not listed as failing.

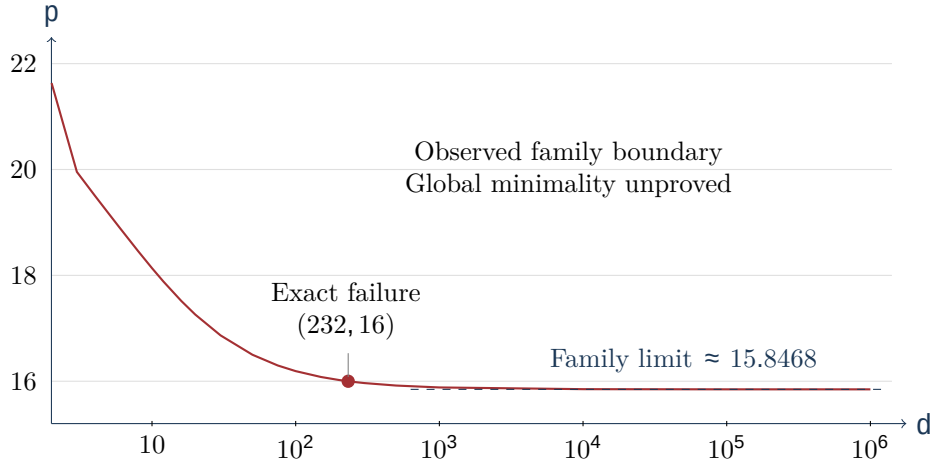


Figure 2: The numerical upper-failure family on a logarithmic dimension axis. The connecting line guides the eye; it is not a global threshold theorem. The dimension-232 mark has a separate exact finite certificate.

Within this family the continuous multiplicity parameter crosses $p = 16$ at dimension approximately 231.60314: the optimized branch is positive at 231 and negative at 232. Other families below 232 are not excluded. A separate exponential limiting-law calculation gives onset 15.84681787, consistent with the large finite-dimensional values. There is no proof that this is $\lim_d p_{\min}(d)$, that this limit exists, or that every dimension is attractive through 15.

The compact follow-up searches comprised 50 cluster-cap cases (all integers 7 through 16 at $d = 3, 10, 100, 1000, 10^6$), 96 grouped regular-fan cases, 18 Gaussian/exponential limiting-law cases, and 24 asymmetric gamma/Gaussian cases, together with the boundary continuations. Fan tests included the individual transverse cubic moments. The general three-exponential search recovered the same limiting branch; the other tested families did not lower it. Selected families and local optimization cannot establish a global minimum or justify a dimension-specific formula.

12 Limiting centers, related mass models, and triangle Hearts

As $p \downarrow 1$, atomic power centers approach geometric medians when the limiting median is unique; otherwise limiting minimizers lie in the median set. On a triangle the Fermat construction is piecewise, depending on its angles. The solid counterpart is a volume geometric median, a different measure problem. No global hull-attractivity result is claimed at $p = 1$, outside the standing domain $p > 1$. At $p = \infty$ both supports lead to minimum-enclosing-ball centers, and the acute circumcenter branch already has indefinite response. For $C = (q, r)$ with $r > 0$,

$$O(q, r) = \begin{matrix} \square \\ 0, \frac{q^2 + r^2 - 1}{2r} \end{matrix}^{\square}, \quad \det \text{Sym } D_C O = -q^2/(4r^2) < 0 \quad (q \not\equiv 0).$$

The exact finite large-power pair gives a separate quantitative realization.

Uniform boundary or skeletal mass is a further model. For the complete m -skeleton of a proper d -simplex, $C_{d,m} = \int_F \text{Vol}_m(F) c_F / \int_F \text{Vol}_m(F)$, where c_F is a face centroid. The extremes $m = 0, d$ are the vertex centroid. Moving a vertex changes both face locations and face masses, producing an extra redistribution term in the derivative. For triangles the wire center is $X(10) = ((b+c)A + (c+a)B + (a+b)C)/(2(a+b+c))$, an attractive rule in the prior theory. For the tetrahedron $(0, 0, 0), (1, 0, 0), (0, 1, 0), (3, -2, 1/2)$, both edge and face mass centers have certified indefinite symmetric responses to movement of the second vertex. They are not the

uniform-solid hull-power model, even at quadratic power, and cannot supply counterexamples to (2).

The original finite triangle-center comparison also has a precise scope [1, 2]. Let $M_- = M_{p_-}$ and $M_+ = M_{p_+}$. The fixed **B46** family retains these two maps and 44 rational maps (**R44**). In the fixed **B76** reference, neither endpoint is universally contained in the convex hull of the other 75 members. Certified support separations occur at side triples (4165, 5213, 1074) for M_- , with gap > 26.18 , and (268, 499, 321) for M_+ , with gap > 4.99 . The reported containment of all 1,142 proved native ETC maps in conv R44 makes the same witnesses separate these endpoints from that known native hull. Those are results for the specified finite reference and its certificate archive, not exposedness on the unknown maximal Heart, nor a proof that **B46** is minimal or maximal. The two endpoint values are also not proved to contain the whole curve $p \sqsupset M_p(P)$ in their convex hull.

The present hull-power maps on $[p_-, 16] \sqcup \{20, 21\}$ supply additional analytic, similarity-equivariant attractive triangle maps. Fixed convex combinations preserve these properties. Whether any of their values enlarge **B46** or are redundant has not been established. Also, convex combinations of center maps do not equal intermediate members of a nonlinear power family; they cannot replace the continuous exponent certificate.

13 ETC neighbors and their independent attractivity

This section records a new comparison with the Encyclopedia of Triangle Centers (ETC), using the numerical fingerprint method from the author’s **Physically Inspired Triangle Centers (PITC)** project [16, 15]. The ETC definitions retain the authorship of Clark Kimberling and the Encyclopedia’s contributors [17]. All comparison results, response investigations, and certificates in this section are attributed to Ismail Hammoudeh, with extensive use of ChatGPT acknowledged.

13.1 Comparison without a target barycentric formula

For a target rule Z , solve its Cartesian position on each of 100 training triangles P_k . Let D_k be the longest side. The positional score of an ETC rule X_j is

$$E_{\text{train}}(Z, X_j) = \frac{1}{100} \sum_{k=1}^{100} \frac{\|Z(P_k) - X_j(P_k)\|^2}{D_k^2}^{1/2}. \quad (49)$$

The triangles are PITC’s fixed equal-area grid in triangle-angle shape space. A separate 40-triangle holdout supplies validation RMS and maximum errors; it is not used to rerank candidates. Thus no symbolic barycentric function for M_p or U_p is needed. “Nearest” means nearest under this specified finite fingerprint, not nearest for every individual triangle.

The supplied snapshot has 72,807 records. All were visited, but only 70,853 have complete finite evaluable fingerprints on the training grid. The other 1,954 are excluded for unsupported symbols or nonfinite values. Unused zero-filled cache rows are not candidates. The result is therefore a ranking within this evaluable snapshot, not a claim about every ETC entry currently available online. Its database SHA256 is

ace501d40dd83785f89b4de8e4ddcc0c5081d82de82e1baf37816d10e547cbfa.

We compare 33 selected powers in $1 < p < 25$, including $65/64$, p_- , 21.63, and 24.99. On the 140 training and validation shapes, all 4,620 pairs of power-center solves resolve. Hull quadrature orders 40 and 64 differ by at most 3.80×10^{-12} times the triangle diameter on this data set. This is a floating-point consistency check, not a rigorous integration error certificate. For obtuse triangles near $p = 1$, atomic solves use a generalized Weiszfeld majorization before Newton iteration to handle centers very close to a vertex.

At $p = 2$, both target rules are exactly $X(2)$, the centroid. At $p = 4$, the leading atomic match is $X(15810)$, with training and validation RMS errors 0.1257% and 0.1322% of diameter; the leading hull match is $X(52788)$, with errors 0.2038% and 0.2080%. Proximity at the other powers establishes neither an identity nor a barycentric formula for the target. The full scores and maximum errors are distributed with `explorer/data/etc-matches.csv`.

13.2 A separate attractivity investigation

An ETC rule has no power parameter here. Its response status belongs to the rule and does not change when it is shortlisted at a different p . For every selected formula, forward automatic differentiation computes the three 2×2 vertex Jacobians. We screen their symmetric parts on 983 proper triangles: the 140 fingerprint triangles, 700 seeded random shapes with apex coordinates in $[-0.8, 1.8]$ and logarithmically distributed heights in $[0.002, 3]$, and 143 thin, obtuse, right, and isosceles family members. All 85 included formulas, including three additional baseline rules, are finite on all these shapes.

Negative differential responses are converted into finite motions. Rational interval arithmetic then verifies proper before/after triangles, nonzero homogeneous weight sums, and a strictly negative upper endpoint for $h \cdot (X(P_{i,h}) - X(P))$. Square roots are enclosed using integer square root on a rational grid with 45 decimal places; subsequent arithmetic uses exact fractions. There are 34 certified failures in the full formula set, of which 33 are actual shortlisted neighbors; the extra failure is baseline $X(6)$.

Among the 82 distinct shortlisted neighbors, 33 are therefore proved nonattractive, 3 are proved attractive by the arguments below, and 46 have no sampled failure but no all-shape proof in this investigation. Small nonnegative sampled eigenvalues do not certify a positive lower bound over all shapes. The response CSV and exact interval records are `explorer/data/etc-attractivity.csv` and `explorer/data/etc-attractivity-certificates.json`.

13.3 An algebraic counterexample for a hull neighbor

Proposition 13.1. The ETC rule $X(7934)$ is not vertex-attractive, although it is the leading fingerprint match for the universally attractive triangle hull-power center U_{12} .

Proof. The ETC barycentrics of $X(7934)$ are the cyclic values of

$$f(a, b, c) = 2b^4 - b^2c^2 + 2c^4.$$

Let $A = (0, 0)$, $B = (1, 0)$, $C = (1/2, t)$ and put $q = 1/4 + t^2$. The weights at A, B, C are respectively $2q^2 - q + 2$, $2q^2 - q + 2$, and $3q^2$. Differentiate after moving C horizontally. The total weight has derivative zero at this symmetric shape, and the horizontal response is

$$\frac{\partial X_x(7934)}{\partial C_x} = \frac{3q^2 - 4q + 1}{7q^2 - 2q + 4} = \frac{(3q - 1)(q - 1)}{7q^2 - 2q + 4}. \quad (50)$$

It is negative whenever $1/3 < q < 1$, equivalently $1/12 < t^2 < 3/4$. At $t = 5/8$, $q = 41/64$ and the derivative is $-1357/22903$. There is also an exact finite witness: with $h = (1/1000, 0)$ applied to C , direct rational evaluation of the squared side lengths gives

$$h \cdot \square X(7934; A, B, C + h) - X(7934; A, B, C)^\square = -\frac{82823496093}{1397892175783000000} < 0.$$

Both endpoint triangles are proper. The fingerprint match at $p = 12$ has training RMS 1.2287% and holdout RMS 1.1196% of diameter. The independent all-shape proof for U_{12} remains valid. $\square$

At $p = 14$ the three hull neighbors $X(7934)$, $X(7911)$, $X(32578)$ all have certified failures. At $p = 16$ the three are $X(7911)$, $X(25682)$, $X(7934)$, again all failing. At $p = 20, 21$, and 21.63 , the three are $X(25682)$, $X(7911)$, $X(7910)$, all failing. In particular, the proved hull endpoints 16, 20, and 21 cannot transfer attractivity to their positional neighbors. The branch estimate 21.63336 supplies no such implication either.

13.4 Positive algebraic certificates for two neighbors

Proposition 13.2. The ETC rules $X(18236)$ and $X(56203)$ satisfy finite vertex attractivity for every motion with proper triangle endpoints. In particular, $X(18236)$ is an attractive leading neighbor of M_7 , even though $7 > 4 + 2\sqrt{2}$ and M_7 is not universally attractive.

Proof. Their first homogeneous barycentric polynomials are

$$\begin{aligned} f_{18236}(a, b, c) &= a(a - b - c)g(a, b, c), \\ g(a, b, c) &= a^3b - a^2b^2 - ab^3 + b^4 + a^3c - 2a^2bc + 3ab^2c + 2b^3c \\ &\quad - a^2c^2 + 3abc^2 - 6b^2c^2 - ac^3 + 2bc^3 + c^4, \\ f_{56203}(a, b, c) &= a(a - b - c)(2a + 2b + c)(2a + b + 2c). \end{aligned}$$

Here $a = |B - C|$, $b = |C - A|$, $c = |A - B|$. The following reduction and its exact coefficient lists provide a finite sufficient certificate. For either f , write

$$W_A = f(a, b, c), \quad W_B = f(b, c, a), \quad W_C = f(c, a, b), \quad T = W_A + W_B + W_C.$$

Place $A = (0, 0)$, $B = (c, 0)$, $C = (x, y)$, so that $x = (b^2 + c^2 - a^2)/(2c)$ and $y^2 = b^2 - x^2$. Define polynomials

$$U_k = c \left[(\partial_k W_B)T - W_B \partial_k T \right], \quad L_k = (\partial_k W_C)T - W_C \partial_k T, \quad k = a, b.$$

Set $z = b^2 + c^2 - a^2$, $\tau = b^2 - c^2 - a^2$, $Y = 4b^2c^2 - z^2$, and

$$\begin{aligned} N_x &= 4abc^2TW_C + 2bcU_a\tau + 2acU_bz + bzL_a\tau + az^2L_b, \\ N_y &= 4abc^2TW_C + Y(bL_a + aL_b), \\ N_o &= 2bc^2U_a + 2ac^2U_b + bc(z + \tau)L_a + 2aczL_b. \end{aligned}$$

Direct differentiation of $X_x = (cW_B + xW_C)/T$, $X_y = yW_C/T$ gives, for $K = \text{Sym } D_C X$ and $F = 4abc^2T^2$,

$$K_{xx} = N_x/F, \quad K_{yy} = N_y/F, \quad K_{xy} = yN_o/F, \quad \det K = \frac{4c^2N_xN_y - YN_o^2}{4c^2F^2}.$$

Substitute $a = v + w$, $b = w + u$, $c = u + v$. Every proper triangle corresponds to $u, v, w > 0$. Exact polynomial expansion gives strictly positive coefficients for every nonzero term of N_y , of $4c^2N_xN_y - YN_o^2$, and of T^2 . The respective numbers of terms are

Rule	N_y	determinant numerator	T^2
$X(18236)$	95	363	61
$X(56203)$	60	266	36

Consequently $T \equiv 0$, $K_{yy} > 0$, and $\det K > 0$ on proper triangles. One positive diagonal and a positive determinant imply $K \succ 0$; no separate positive-coefficient expansion of N_x is needed. Cyclic relabeling proves the same property for each moved vertex. The exact expansions are distributed in [explorer/data/etc-positive-certificates.json](#) and regenerated

by `research/etc_matching/prove_positive.py` , which uses only integer and rational sparse polynomial arithmetic.

The T^2 expansions also contain a positive term on each open boundary face $u = 0$, $v = 0$, or $w = 0$. Thus the formulas remain defined and continuous at every distinct collinear triangle, and their response matrices there are positive semidefinite by continuity. Integrating the local response along a single-vertex segment which avoids coincidences gives the finite inequality. A segment through a coincidence can be approximated by segments avoiding it, keeping its two proper endpoints arbitrarily close to the original endpoints. Continuity at those endpoints then gives the finite inequality in the limit. No prescribed value at an exactly coincident input is asserted for these two ETC formulas. $\square$

The leading $p = 7$ atomic shortlist is $X(18236)$, $X(30624)$, $X(46844)$. The first rule has the positive proof just given, while the latter two have exact finite failures. The leading match has holdout RMS 1.1433% of diameter. Thus failure of M_p above its universal interval does not force failure of its neighboring ETC rules. $X(56203)$, for example a hull neighbor at $p = 8$, supplies a second positive all-shape certificate. Together with the centroid's exact response $D_{v_i} X(2) = 1/3$, these are the three accepted positive neighbor results. The remaining numerical passes stay unproved.

13.5 All selected powers and the explorer

In the following table the three ETC numbers are ordered by training RMS. Superscript **A** means proved attractive (the proper-endpoint scope above applies to 18236 and 56203); **F** means certified finite failure; **?** means no sampled failure and no all-shape proof. These are statuses of ETC rules, not new statements about the corresponding target powers.

The free browser explorer at ismaa3iil.fyi/power-centers-explorer/ now offers all 33 comparisons, overlays of either or both shortlists, training, validation, and current-triangle distances, and separate response-status labels. “Show failure” loads a certified triangle; the stored motion can be applied and reset. Displayed positions and local response values remain floating-point diagnostics; the downloadable exact records support the proof labels. Arbitrary powers between the sampled values do not receive interpolated rankings. The triangle range is $[1.01, 24.99]$ and the tetrahedron range remains $[p-, 19.95]$. ETC comparisons concern triangles only. The explorer uses free local JavaScript tools and includes only selected formulas, not the full database. Proximity never changes the attractivity claims proved in earlier sections, nor certifies all $p < 21.63$ for U_p .

14 Reproducibility and the status of the proofs

The analytic arguments and the finite certificate data have different roles. The common dimension-independent extension is proved by the exponential-moment and matrix estimates; its small constant checker verifies the displayed rational margins. The triangle theorem additionally depends on complete finite covers and exact arithmetic replay. Floating calculations propose boxes, center guesses, and preconditioners. They make no acceptance decision in the final proof. Exact finite counterexamples include both integration error and residual-to-center error. Numerical searches guide the investigation and carry no all-shape positivity conclusion.

All exact replay programs use ordinary Python 3 with its standard library. They must run without `-O`, because assertions are part of the verification. The `-S` option demonstrates independence from installed packages. Construction and exploratory numerical searches require NumPy. Useful commands from the workspace root are:

```
python -S research/verify_positive_bound.py
python -S research/triangle8_interval.py -replay
python -S research/triangle10_interval.py -replay
```

Table 5: ETC fingerprint neighbors and independent response status.

p	Atomic neighbors			Hull neighbors		
1.01	54788 ^F	11544 ^F	1996 ^F	55759 [?]	63120 [?]	31312 [?]
65/64	60083 ^F	11544 ^F	1996 ^F	55759 [?]	55758 [?]	63120 [?]
1.05	7272 ^F	60258 ^F	60617 ^F	55758 [?]	4798 [?]	51126 [?]
1.1	42321 ^F	56929 ^F	4059 ^F	4798 [?]	51126 [?]	55757 [?]
p-	61013 ^F	61027 ^F	62919 ^F	60644 [?]	55756 [?]	55755 [?]
1.25	62921 ^F	30380 ^F	57722 ^F	55754 [?]	6707 [?]	55755 [?]
1.5	60096 [?]	60204 [?]	36511 [?]	55750 [?]	55751 [?]	62940 [?]
2	2 ^A	51553 [?]	53004 [?]	2 ^A	51553 [?]	53004 [?]
2.5	7944 [?]	5599 [?]	17250 [?]	61001 [?]	19872 [?]	31253 [?]
3	69642 [?]	3305 [?]	54357 [?]	31268 [?]	15894 ^F	52706 [?]
4	15810 [?]	56508 [?]	29492 [?]	52788 [?]	46225 [?]	7914 [?]
5	7928 ^F	15810 [?]	32578 ^F	7937 [?]	5506 [?]	5316 [?]
6	46844 ^F	18236 ^A	50219 ^F	7853 [?]	31246 [?]	24954 [?]
7	18236 ^A	30624 ^F	46844 ^F	7853 [?]	7935 [?]	24707 [?]
8	30624 ^F	18236 ^A	40175 ^F	24707 [?]	7935 [?]	56203 ^A
9	40175 ^F	30624 ^F	15855 ^F	7831 [?]	24707 [?]	56203 ^A
10	40175 ^F	15855 ^F	59646 ^F	7831 [?]	29492 [?]	56508 [?]
11	40175 ^F	15855 ^F	59646 ^F	7831 [?]	29492 [?]	56508 [?]
12	40175 ^F	15855 ^F	31594 ^F	7934 ^F	15810 [?]	56508 [?]
13	40175 ^F	15855 ^F	31594 ^F	7934 ^F	7911 ^F	15810 [?]
14	40175 ^F	15855 ^F	6554 ^F	7934 ^F	7911 ^F	32578 ^F
15	40175 ^F	15855 ^F	6554 ^F	7911 ^F	7934 ^F	25682 ^F
16	40175 ^F	6554 ^F	15855 ^F	7911 ^F	25682 ^F	7934 ^F
17	40175 ^F	6554 ^F	15855 ^F	25682 ^F	7911 ^F	7934 ^F
18	40175 ^F	6554 ^F	15855 ^F	25682 ^F	7911 ^F	7910 ^F
19	6554 ^F	40175 ^F	15855 ^F	25682 ^F	7911 ^F	7910 ^F
20	6554 ^F	40175 ^F	15855 ^F	25682 ^F	7911 ^F	7910 ^F
21	6554 ^F	40175 ^F	5574 ^F	25682 ^F	7911 ^F	7910 ^F
21.63	6554 ^F	40175 ^F	5574 ^F	25682 ^F	7911 ^F	7910 ^F
22	6554 ^F	40175 ^F	5574 ^F	25682 ^F	7911 ^F	7910 ^F
23	6554 ^F	5574 ^F	40175 ^F	25682 ^F	7911 ^F	7910 ^F
24	5574 ^F	6554 ^F	9713 ^F	25682 ^F	7911 ^F	7910 ^F
24.99	5574 ^F	6554 ^F	9713 ^F	25682 ^F	40175 ^F	7911 ^F

```
python -S research/triangle12_interval.py -replay
python -S research/triangle_extended_integer.py -power 16 -workers 4
python -S research/triangle_extended_integer.py -power 20 -workers 4
python -S research/triangle_extended_integer.py -power 21 -workers 4
python -S research/triangle_power_integer_replay.py -workers 4
python -S research/triangle_extended_integer.py -power 16 -continuous -workers 4
python -S research/verify_dimension232.py
python -S research/downward_closure_countercheck.py
```

The $p = 16$ endpoint replay checks 74,996 leaves, depth 20. The continuous $[10, 16]$ replay checks 3,042,242 leaves, depth 28; its completed eight-worker run took approximately 7,939 seconds on the investigation machine. Worker counts affect runtime, not the mathematical tests. Streaming workers retain at most 128 proposals at a time, while the parent audits the full cover and verifies each worker input hash and record count. Independent moment and interval-primitive cross-checks support the implementation; none substitutes for the complete replay.

Four principal certificate SHA256 values are

```
Endpoint  $p = 16$ :
52ee00ac65cb6601a88e5c144cd2fb8e238b3f624a7da87d434ee4c87ec98c6e
Continuous  $[10, 16]$ :
59046abff6f15c663f2c8a5b287f33565f9187915458f714c35b1cbe977fb0d6
Endpoint  $p = 20$ :
d18f39bf441fb16b9b7465f4264eceb60281a9cf7ad1ccb3245f764aec660fab
Endpoint  $p = 21$ :
20b338bf9d4fbc4027e970297c2acd85e8a75bf374844eb44eebd609f2499dbc
```

The focused distribution is `Triangle16_ContinuousInterval_Certificates_2026-10-03.zip`. It passed archive integrity and file-hash checks and independent validators run from an extracted copy. Its SHA256 is

```
bb545adce451cee7f76932b350c9a292997359902ac6d8a78272641ef9e8e4a1.
```

The full investigation distribution is `Hull_Power_Investigation_Triangle16_Interval_2026-10-03.zip`; its verified SHA256 is

```
152a6816eaac408c3ed20ac5d72f815fad01dc8ec5ae7e47633f1ef4354084eb.
```

The later focused distributions are `Triangle20_LowPower_Certificates_2026-10-03.zip` and `Triangle21_Endpoint_Certificates_2026-10-03.zip`. The final-statement records pin endpoint source and replay hashes. The dimension-232 certificate separately pins its exact-moment dependencies. No completed complete-cover replay for $[16, 21]$ is included among the accepted records as of this revision. Partial continuation work is not counted as an interval theorem.

For this rewritten article, an editorial audit compares table entries, rational margins, and cited paths against the accepted records. It does not rerun the multi-million-box proofs. Analytic structural results have their proofs above and in the cited notes; the associated floating-point diagnostics are labeled separately. Existing certificate archives are preserved. This article supplies neither a sharp cutoff nor a global dimension formula.

15 Remaining mathematical questions

The central triangle problem is to connect the continuous interval through 16 with the separately certified endpoints 20 and 21. A complete shape/exponent certificate over $[16, 21]$ or a proof of all-shape downward closure would do so. The exact crossing identity (48) isolates a sufficient sign inequality; its universal sign remains open. Fixed-triangle and fixed-motion no-re-entry are false and cannot supply that proof. Existence, uniqueness, and eventual failure of the projective families do not establish persistence from the first failure.

The observed triangle branch near 21.63336 needs a global exclusion of other shapes and branches before it can be called sharp. The lower hull-power interval $(1, p_-)$ also remains unclassified. In particular, the local certificates and positive search evidence at 65/64 do not settle its all-shape question; atomic necessity does not transfer.

For tetrahedra, all-shape certificates at 16 or 18 would convert strong search evidence into theorems. Further proper higher-dimensional families, all-vertex response tests, and sharper analytic common bounds are needed to identify dimension dependence globally. The dimension-232 witness improves the upper bound for $d_{\min}(16)$, but a smaller failure dimension is not excluded. For each integer 7 through 15, no finite failure dimension is established. The observed limit 15.8468 belongs to the examined family and is not a dimension-independent sharp endpoint.

Connectedness of the **all-shape** attractive sets, upward persistence of the existence of some bad simplex, and nesting across intrinsic dimensions all remain open. These are distinct from the disproved fixed-shape no-re-entry assertion. The small common extension δ reflects conservative estimates rather than a proved near-band obstruction. Finally, a certified support comparison of hull-power triangle maps with B46 is separate from the attractivity problem and remains to be carried out.

A Source and certificate inventory

All paths below are relative to the workspace root; the standalone article has no compilation dependency on them.

Atomic classification, radial theory, continuum consistency

bundle/simplex_integral_power_centers_handoff_2026-10-01/paper/Attractive_Centers_III_v2.tex

Transfer, proper analyticity, quartic moments, large-power pair

bundle/simplex_integral_power_centers_handoff_2026-10-01/POWER_CENTERS_AND_UNIFORM_SIMPLEX_HANDOFF.md

Concrete atomic and skeletal witnesses

bundle/simplex_integral_power_centers_handoff_2026-10-01/existing_certificates/verify_interval_certificates.py

Every-dimensional positive extension

research/ALL_SHAPE_AND_DIMENSION.md
research/verify_positive_bound.py

Triangle 8, 10, 12, 16 endpoints

research/triangle8_interval.py
research/triangle10_interval.py
research/triangle12_interval.py
research/triangle_extended_integer.py

Continuous triangle intervals

research/triangle_power_integer_replay.py
research/triangle_extended_integer.py
research/triangle_commutator_response.py

Triangle 20, 21 endpoints and low power

research/TRIANGLE20_AND_LOW_POWER.md
research/TRIANGLE21_ENDPOINT.md
research/certify_triangle65_64.py
research/verify_atomic65_64.py

Endpoint replay records

research/triangle20_endpoint_integer_replay.json
research/triangle21_endpoint_integer_replay.json

Triangle finite failures
research/verify_rational_power.py
research/verify_p22_finite.py

Tetrahedron and proper dimensions 5, 31
research/verify_tetra20.py
research/verify_dimension5_p20.py
research/verify_dimension31.py

Proper dimensions 12, 302
research/verify_cluster_caps_finite.py
research/cluster_caps_finite_certificates.json

Proper dimension 232
research/verify_dimension232.py
research/dimension232_certificate.json

Fixed-triangle and fixed-motion re-entry
research/DOWNWARD_CLOSURE_COUNTERCHECK.md
research/downward_closure_countercheck.py

Downward closure, disk test, projective families
research/EXPONENT_DOWNWARD_CLOSURE.md
research/DOWNWARD_CLOSURE_PROJECTIVE_REDUCTION.md

Crossing diagnostics
research/validate_downward_projective_reduction.py
research/downward_crossing_probe_roots.json

Dimension/power profile
research/DIMENSION_FAILURE_PROFILE.md
research/dimension_failure_profile.json
research/dimension16_improvement.json

Triangle, tetrahedron, higher-dimensional searches
research/global_triangle_search.py
research/tetra_global_search.py
research/higher_proper_search.py
research/cluster_caps_search.py

ETC fingerprint comparisons and neighbor proofs
research/etc_matching/build_inputs.py
research/etc_matching/compute_centers.mjs
research/etc_matching/rank.py
research/etc_matching/screen_attractivity.mjs
research/etc_matching/certify_neighbors.py
research/etc_matching/prove_positive.py

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- [2] Ismail Hammoudeh, **Power Centers and the Uniform-Simplex Modification**, research hand-off, 1 October 2026. Local source identified in the appendix.
- [3] Ismail Hammoudeh, **Hull-Power-Centers: First Attractivity Investigation**, 1 October 2026, research/INVESTIGATION.md

- [4] Ismail Hammoudeh, **All-Shape Positivity and Dimension Dependence of Hull-Power-Centers**, 1 October 2026, with continuation notices through 3 October, [research/ALL_SHAPE_AND_DIMENSION.md](#)
- [5] Ismail Hammoudeh, **Triangle $p = 8$ Certificate and Proper Higher-Dimensional Failures**, 2 October 2026, [research/TRIANGLE8_AND_GLOBAL_SEARCH.md](#)
- [6] Ismail Hammoudeh, **All-Shape Triangle Attractivity at $p = 10$** , 2 October 2026, [research/TRIANGLE10_ALL_SHAPE.md](#)
- [7] Ismail Hammoudeh, **Triangle Hull-Power Centers: $p = 12$ and a Continuous Exponent Interval**, 2 October 2026, with a later-result notice on 3 October, [research/TRIANGLE12_AND_CONTINUOUS_INTERVAL.md](#)
- [8] Ismail Hammoudeh, **Triangle Hull-Power Centers at $p = 16$ and over a Continuous Interval**, 3 October 2026, [research/TRIANGLE16_AND_CONTINUOUS_INTERVAL.md](#), and the completed endpoint and continuation integer replay records.
- [9] Ismail Hammoudeh, **Triangle Hull-Power Centers at $p = 20$ and $p = 65/64$** , 3 October 2026, [research/TRIANGLE20_AND_LOW_POWER.md](#), and exact endpoint, local low-power, and atomic certificate records.
- [10] Ismail Hammoudeh, **All-Shape Triangle Hull-Power Attractivity at $p = 21$** , 3 October 2026, [research/TRIANGLE21_ENDPOINT.md](#), with the completed independent integer replay.
- [11] Ismail Hammoudeh, **Does an Attractive Upper Exponent Force Every Lower Exponent above Four?**, revised 5 October 2026, [research/EXPONENT_DOWNWARD_CLOSURE.md](#)
- [12] Ismail Hammoudeh, **Fixed-Triangle No-Re-Entry Is False**, 4–5 October 2026, [research/DOWNWARD_CLOSURE_COUNTERCHECK.md](#), with polynomial certificate and asymptotic proof.
- [13] Ismail Hammoudeh, **Downward Closure: A Projective Comparison and a Disk Criterion**, revised 5 October 2026, [research/DOWNWARD_CLOSURE_PROJECTIVE_REDUCTION.md](#)
- [14] Ismail Hammoudeh, **Numerical Failure Dimensions and Powers for Hull-Power Centers** 5 October 2026, [research/DIMENSION_FAILURE_PROFILE.md](#), with the exact dimension-232 certificate and numerical search records.
- [15] Ismail Hammoudeh, **ETC Neighbors of Power Centers and Hull-Power Centers: Fingerprint Rankings and Independent Attractivity Certificates**, 6 October 2026, [research/ETC_NEIGHBORS.md](#), with the numerical rankings, exact finite failures, and positive polynomial coefficient records.
- [16] Ismail Hammoudeh, **Triangle Centers Inspired by Physics** PITS project, 3 October 2026; numerical ETC comparison protocol and supplied database snapshot. [Author's research website](#).
- [17] Clark Kimberling and contributors, **Encyclopedia of Triangle Centers** Official online encyclopedia, including the entries X(7934), X(18236), and X(56203); accessed 6 October 2026.
- [18] Ben Andrews, Pengfei Guan, and Lei Ni, **Flow by Powers of the Gauss Curvature**, *Advances in Mathematics* **299** (2016), 174–201. doi:10.1016/j.aim.2016.05.008.